Timing is Everything: Parental Professional Connections and Labor Market Entry

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Abstract

I study the effect of parents' former coworkers on young workers in Israel. Comparing firms where a parent's coworker was present at the worker's entry year to firms where the coworker had left shortly before or arrived shortly after, I isolate the causal effect of these connections. Workers are 3.3 times more likely to find their first job at a firm with an active connection. Effects are stronger for disadvantaged groups and where match quality matters less. Connected workers earn higher entry wages but experience slower wage growth. These patterns are consistent with connections primarily transmitting information about vacancies.

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1 Introduction

In his influential hypothesis "The Strength of Weak Ties," Granovetter (1973, 1983) argues that weak ties—the network of acquaintances—play a crucial role in the dissemination of information by acting as bridges between different social circles. In the context of the labor market, Granovetter suggests that these weak ties are mainly important for "knowing about appropriate job openings at just the right time" (Granovetter 1983, pp. 202). The modern theoretical job-search literature often assumes such a mechanism: that different labor-market participants learn of new vacancies at different rates, and that social connections can increase that rate (Mortensen and Pissarides 1994; Burdett and Mortensen 1998; Calvó-Armengol 2004; Fontaine 2008).

Despite the theoretical prominence of this idea, empirical evidence on the causal effects of weak ties and on the importance of the information channel in job search remains limited. In this paper, I examine the labor market effects of weak ties for young workers entering the labor market. I focus on a particular type of weak tie that is especially relevant for young workers who lack their own professional network: their parents' former colleagues.

To identify the causal effect of weak ties, I develop a novel identification strategy that leverages the timing of the contact's presence at the firm. In particular, I compare firms where the employee had *active* links in the labor-market entry year ("weak connections") with firms with *non-active* links, where the contact had left shortly before or joined shortly after the labor-market entry year ("phantom connections"). The key identifying assumption is that, absent the effect of connections themselves, there are no systematic differences between firms with weak and phantom connections that affect workers' employment probability.

I implement this strategy using matched employer-employee administrative records covering the entire Israeli workforce from 1991 to 2015, linked to the Israeli Population Registry, which provides demographic information and parent–child identifiers. The analysis sample comprises Israelis aged 22–27 who found their first stable job at a firm with 5–500 workers between 2006 and 2015.

Using this data, I show that firms with weak and phantom connections are similar across a variety of characteristics such as sector, location, and hiring trends, and that the effect is unlikely to result from endogenous separations of the contacts—supporting the identification assumption stated above.

I find substantial effects of weak parental ties on hiring. For workers with a single contact, having a weak connection increases the probability of working at a particular firm by 0.052 percentage points (about one-twentieth of a percent) relative to no connection, compared to only 0.012 percentage points for a phantom connection, a statistically significant difference of 0.040 percentage points. This implies that a weak connection increases the probability of working in a firm by 3.3 times compared to a phantom connection. The effects are substantially larger for multiple connections: workers with two weak connections are approximately 7 times more likely to work at connected firms, while those with three or more connections are about 21 times more likely. The probability of a worker starting at a particular firm decreases significantly the year after the connection is ended.

Connections are more effective when formed at smaller firms, for longer durations, and more recently. The effects are stronger when the child, parent, and parent's coworker share characteristics such as gender or ethnicity. Furthermore, the effects are more pronounced for individuals from the Arab minority, less-educated workers, workers from lower socioeconomic backgrounds, and during periods of high unemployment. The effects are also more pronounced for males than for females.

Importantly, I find that connection effects are stronger in firms where match quality is less important—firms with lower average match duration, lower variance in match duration, and smaller wage match effects. If connections mainly helped workers find better matches, we would expect stronger effects in environments where match quality matters more. Instead, connections appear most valuable in facilitating access to jobs where worker-firm pairing is relatively homogeneous and less consequential for outcomes.

Turning to wage and career outcomes, weak connections are associated with higher entry wages than phantom connections: 1.5 percent for one connection, 3.1 percent for two, and 5.6 percent for three or more. However, workers with weak connections experience slower wage growth, and the initial wage premium dissipates over time. Workers with connections to their first firm also have modestly higher job tenure there, approximately one additional month for those with a single connection. Examining longer-term earnings, I find that cumulative five-year discounted earnings advantages are concentrated among workers with multiple connections: essentially zero for single connections, rising to about 6 percent for those with three or more. Weak connections show no advantage in subsequent job salaries after workers leave their first employer.

These wage and career patterns—a modest entry-wage premium, slower wage growth, and no advantage in subsequent jobs—are consistent with connections primarily reducing search frictions rather than improving match quality. The finding that connection effects are stronger in settings where match quality is less important and for workers at the low-skill end of the labor market—where the returns to careful screening for unobserved match quality are smaller, and information is correspondingly more valuable—reinforces this interpretation, which aligns with Granovetter (1973)'s hypothesis that weak ties are mainly valuable for learning about job openings.

This paper makes two main contributions. First, it is the first to study the labor market effects of parents' colleagues' networks, offering a novel causal identification strategy that leverages the timing of contacts' movements between firms. While existing literature finds that strong ties (where parents work) increase a child's probability of working there (Corak and Piraino 2011; Kramarz and Skans 2014; Stinson and Wignall 2018; Staiger 2021; Eliason et al. 2023), evidence for the impact of weak parental ties remains limited. Previous studies found no impact for different types of weak parental connections, such as parents of high-school classmates (Kramarz and Skans 2014) or parents' high-school classmates (Plug et al. 2018).[1] The positive effect I find for parents' past coworkers' networks aligns with literature showing the importance of coworker networks for workers' own labor market outcomes (Granovetter 1974; Cingano and Rosolia 2012; Glitz 2017; Caldwell and Harmon 2019).[2]

Second, the paper aims to better understand the mechanism through which social connections affect workers. As mentioned above, the theoretical job-search literature often assumes that different labor-market participants learn of new vacancies at different rates—for example, employed workers relative to unemployed workers (Mortensen and Pissarides 1994; Burdett and Mortensen 1998), or connected and non-connected workers (Calvó-Armengol 2004; Fontaine 2008). The combination of evidence summarized above is consistent with this information channel about firm vacancies, as opposed to other channels in which connections affect match quality or productivity (Montgomery 1991; Beaman and Magruder 2012; Bolte et al. 2020).

The closest paper to this one is Kramarz and Skans (2014) (KS hereafter), who use Swedish administrative data to study parental connections in young workers' labor-market entry. For comparison with KS and related work, I also report estimates of strong parental connections throughout the paper, and find that effects on hiring, tenure, and wage growth are quantitatively comparable to theirs, with both papers finding effects to be larger for more disadvantaged workers. This paper extends KS in two directions: it introduces phantom connections as a new identification strategy that exploits the timing of contacts' movements between firms, and uses this strategy to provide evidence on the information channel in job search. The papers also diverge on the estimates of weak ties: KS find negligible effects of classmates' parents, while I find substantial effects of parents' past coworkers. This difference suggests that how the weak tie is constructed matters: a parent's past coworker has a professional relationship with the parent and may share information about openings at the

[1] I find that the effect of a parent's past coworker on a child's hiring probability is larger the more recently the parent and coworker overlapped at the firm (see Figure 3), which may help explain why long-dated ties such as parents' high-school classmates show negligible effects.

[2] Glitz (2017) is the earliest contribution examining the role of ex-coworkers in workers' job-to-job transitions.

firm where they currently work, whereas the chain from worker to classmate to classmate's parent contains no professional link at any step.

The rest of the paper is structured as follows. Section 2 describes the data, definitions, and identification strategy. Section 3 presents the regression framework and results on employment effects. Section 4 examines salary and job tenure. Section 5 discusses the findings and their implications. Section 6 concludes.

2 Data and identification strategy

2.1 Data and definitions

I use matched employer-employee administrative records from Israel. The data contain administrative information about the entire Israeli workforce from 1991 to 2015 collected from tax records. The dataset includes person identifiers, firm identifiers, monthly employment indicators for each firm in which a person worked, the yearly salary received from each firm in a year, and the firms' industry.

The employment tax records are merged with the Israeli Population Registry. This dataset covers the full population of Israel. It includes demographic information: date of birth, date of death (if any), sex, ethnic group, country of birth, and date of immigration to Israel. Most importantly for this study, the data include identifiers of each individual's parents, which enables me to link parents and children.

This paper studies the impact of the professional network of parents on the employment and salary of young workers entering the labor market. The paper's primary focus is on weak parental connections, defined as connections between workers and firms where at least one of their parents' past coworkers was employed during the worker's labor-market entry year. Past coworkers are defined as individuals who worked at the same firm as the parent when the child was 12–21 years old. For comparison, I also examine the effect of strong connections, defined as firms where a worker's parent was employed when the child was 12–21 years old.[3]

Following Kramarz and Skans (2014), I define the first stable job as the first job after higher-education graduation (if applicable) that lasts for at least four months during a calendar year and produces total annual earnings corresponding to at least 150% the national average monthly wage. The labor-market entry year is the year the new worker finds her

[3]To correctly measure the impact of weak connections throughout this paper, I compare the treatment (weak connections) and control (phantom connections) groups using the same number of connections for each. Specifically, I separately examine firms with one, two, and three or more connections of each type.

first stable job.[4]

My analysis sample comprises the first stable job of Israelis between ages 22-27 in a 5-500 workers firm who found this job in the years 2006-2015. I exclude workers without any parent who worked in a 5-500 workers firm when they were 12-21 years old. I further exclude immigrants and Ultraorthodox Jews from the sample.[5]

2.2 Summary statistics

Table 1 shows sample sizes and sample means. The new workers' sample—my main analysis sample—includes 220,806 workers, of which 29% are Arabs, 43% are female, and 23% have some college education. The average age at first stable employment is 24, and the average monthly salary is 5,839 NIS, which is equivalent to 1,621 USD (2017 prices).

On average, Jews who enter the labor market earn more at their first job and work at higher paying firms (in terms of AKM pay premiums) than Arabs. Additionally, Jews are connected to higher-paying firms via both strong and weak connections. However, the share of workers who find their first job in a connected firm is higher for Arabs than for Jews. Specifically, 5.9% of Arab workers have at least one weak connection at their first job compared to 2.8% of Jewish workers, and 10.2% of Arab workers have strong connections compared to 5.8% of Jewish workers (Table 1).

Comparing males and females, males earn more at their first job but work at similar-paying firms to females. Likewise, males are connected to firms with slightly lower average rank (in terms of AKM pay premium) than females.[6] Finally, the share of workers who find their first job in a connected firm is higher for males than for females: 4.3% of males have at least one weak connection at their first job compared to 2.6% of females, and 8.6% of males have strong connections compared to 5.0% of females.

To better understand the distribution of connections, I group the firms into five bins using their pay premiums. Figure 1 shows the number of weak connections (with one contact only) and strong connections within each bin of firms for different groups of workers. Panel A shows that, on average, Jews and Arabs have similar numbers of weak connections with firms at the lowest quintile of pay premiums. However, Jews have substantially more weak connections with higher-ranked firms than Arabs, and the gap increases as the firm's rank increases. For strong connections, Panel B shows that Jews have fewer connections with low-paying firms

[4]I check the robustness of the results for two alternative definitions of the labor-market entry year: 1) The year in which the worker is 25 years old, and 2) The year after the worker's graduation year (Figure A7).

[5]See Appendix B for further information about the data construction and the definitions of the variables.

[6]This difference is due to the fact that the participation rate of Arab women in the labor market is much lower than that of Jewish women.

but more connections with high-paying firms compared to Arabs. Overall, the quality of connections (in terms of the pay premium of the connected firms) is better for Jews than Arabs.

Panels C and D show patterns by gender. Females have slightly better weak connections than males, with more connections to higher-ranked firms. For strong connections, females have slightly fewer connections with firms in the bottom quintile but slightly more connections with higher-ranked firms compared to males, though the overall pattern is quite similar across genders.

2.3 Identification strategy

How much more likely is the average worker to work in a connected firm than in an unconnected firm? A naive comparison between connected and unconnected worker-firm pairs might attribute the effect of omitted variables to the estimated impact of connections. There might be several reasons why a worker is more likely to work in a connected firm, even without the impact of connections per se. For example, Galor and Tsiddon (1997) claim that children tend to choose their parents' occupation because of specific human capital transmitted from parents to children. As the parent's colleagues are also likely to have this particular human capital, the child's probability of working at a firm employing one of these parent's colleagues might be high because both have the same specific human capital. Another example is geographical proximity, which might be correlated with connections to specific firms and impact the employment probability.

This paper addresses this potential endogeneity concern by comparing the probability of working in a firm with an *active* social tie ("weak connections") with a firm with *non-active* social connections ("phantom connections"). In particular, it compares the likelihood of working for a firm where one of the worker's parents' past coworkers worked during the labor-market entry year with the likelihood of working for a firm where the contact had worked there within five years *before or after* the worker entered the labor market, but not that year.

The identification strategy relies on the assumption that the timing of parents' past coworkers' movements between firms is orthogonal to the employment outcomes of the children. Under this assumption, there are no systematic differences between firms with active and non-active connections, except for the presence of a contact at the appropriate time at the actively (weakly) connected firm. Therefore, comparing the two types of firms isolates the impact of connections themselves.[7]

[7]This comparison also addresses potential measurement error from using coworkers to define connections, as such error occurs symmetrically in both the treatment and control groups.

To illustrate this, consider again the example from Galor and Tsiddon (1997) about job-specific human capital that parents pass on to their children. Due to this factor, and the fact that the parent's colleague worked in both the parent's firm and the new firm, children might be more likely to work in firms with weak connections even without the true effect of connections. However, this is equally true for phantom-connected firms, in which the parent's colleague also worked. Therefore, comparing the effect of weak ties relative to phantom ties will eliminate this and similar confounding factors.

Note that phantom connections might still affect the probability of working at a firm. A parent's past coworker at a specific firm might provide relevant information to their contact, either through retained knowledge about the firm or through ongoing connections within it. Therefore, the estimates obtained using this identification strategy represent lower bounds of the actual effect.[8]

Note also that the assumption that the timing of contacts' job movements is orthogonal to workers' labor market entry makes much less sense when applied to the parents themselves rather than to their coworkers.[9] For example, a parent might have left the firm because the family moved to a different location, which might itself reduce the probability of the child working at that firm. However, it is harder to think of similar threats related to the timing of coworker movements. Likewise, this assumption is more appealing when studying workers' first job choice compared to endogenous moves during their career. Section 3.5 provides further discussion and empirical validation of the identification strategy.

3 Effects of parental connections on employment

3.1 Regression framework

Building on Kramarz and Skans (2014), the probability that worker i starts working at firm j is:

$$e_{ij} = \phi_{x(i)j} + \sum_{c} \delta^c \cdot D^c_{ij} + \epsilon_{ij}. \tag{1}$$

e_{ij} is an indicator variable equal to one if individual i starts working at firm j. $\phi_{x(i)j}$ is a match-specific effect that captures the propensity that a worker from a given observable

[8] Alternatively, another type of "phantom" connection could include workers who were employed at the same firm as the parents but in non-overlapping years; see Hensvik and Skans (2016) for a similar approach. However, using that specification as the control group assumes that parents' job movements are orthogonal to their children's job decisions, which is a less attractive assumption in the context of this paper.

[9] Kramarz and Skans (2014) compare situations when the parent is present at the firm versus when the parent has recently left the firm in some of their robustness tests.

group $x(i)$ ends up working at a particular firm. Workers' groups include all combinations of ethnicity, gender, education, age, year of first job, and district of residence.[10] D^c_{ij} is an indicator variable capturing whether worker i has connections of type c to firm j. The main specification includes separate indicators for weak and phantom connections with one, two, and three or more connections, an indicator for mixed connections (at least one weak and one phantom connection), and an indicator for strong connections. The parameters of interest that measure the effect of parental connections are the δ^c coefficients. They estimate how much more likely a firm is to hire a new worker with c-type connections than an unconnected worker from the same group.

My identification strategy compares the difference between the estimates of weak and phantom connections to identify the causal effect of weak connections. Specifically, I do not assume that $E(\epsilon_{ij}|D^c_{ij}, x(i) \times j) = 0$. This assumption is not plausible as jobs with some sort of connections are different from jobs without any connections.[11] Instead, I assume that $E(\epsilon_{ij}|D^{w,n}_{ij}, x(i) \times j) = E(\epsilon_{ij}|D^{p,n}_{ij}, x(i) \times j)$ for $n \in \{1, 2, 3+\}$, where n denotes the number of connections. Using this assumption, I can identify the effect of weak connections with n contacts by taking the difference between $\delta^{w,n}$ and $\delta^{p,n}$. Sections 3.4 and 3.5 provide evidence to support this identification assumption.[12]

Direct estimation of equation (1) is computationally infeasible, as it requires one observation per worker-firm pair, amounting to more than ten billion observations. To estimate equation (1), I apply an extended version of the fixed-effects transformation proposed by Kramarz and Thesmar (2013) and Kramarz and Skans (2014).[13]

3.2 Distribution of connections

Figure A2 shows the distribution of the number of connections per worker by connection type. Across all types, the distributions are right-skewed, with most workers having relatively few connections. However, there is substantial heterogeneity across individuals. Overall, workers have about three times more phantom connections than weak connections.

The distribution of connections across past firm sizes (Figure A3) shows that more weak or phantom connections are generated in larger firms. The distribution of connections across current firm size (firms that the new workers have connections to when entering the labor

[10] See Figure A1 for the distribution of groups and workers by group size.

[11] For example, jobs with one connection of each type are geographically closer and more likely to belong to the parent's industry relative to non-connected jobs (Table A4).

[12] For completeness, I also report the coefficients for mixed and strong connections, although I do not have an appropriate control group for them. Therefore, these results should be interpreted with caution. See Appendix C.1 for a formal discussion of the model assumptions and identification.

[13] See Appendix C.2 for further information about this methodology.

market), however, is more uniform (Figure A4). Strong connections (Panel H) display a relatively flat distribution across both past and current firm sizes.

3.3 Regression results

Table 2 presents estimates of the coefficients in equation (1).[14] Each column shows a separate estimate for a different population group based on ethnicity and gender. All estimates of the effect of connections are positive and statistically significant, implying that new workers with any connections to a firm are more likely to work there than workers with similar observable characteristics but no connections to the firm.

The regression results show that the estimate is much larger for weak connections than for phantom connections, and that the effect (the gap between the coefficients of weak and phantom connections) increases with the number of connections. Having one weak connection at the firm increases the probability of working there by 0.052 percentage points relative to someone with no connections. In contrast, one phantom connection increases this probability by only 0.012 percentage points. The difference between weak and phantom connections with one contact is 0.040 percentage points and is statistically significant. The effects are considerably larger for multiple connections: two weak connections increase employment probability by 0.302 percentage points (difference from phantom: 0.263), while three or more weak connections increase it by 1.025 percentage points (difference from phantom: 0.981). For completeness, the table also reports estimates for mixed connections (0.248 percentage points) and strong connections (3.091 percentage points).

To better understand the magnitude of these effects, I calculate the ratio between the likelihood of working in weakly-connected firms and phantom-connected firms. The estimated probability of working in a firm with one weak connection is 3.28 times higher than the probability of working in a firm with one phantom connection for the average new worker (Table 2, column 1). This ratio increases substantially with additional connections: 6.96 for two connections and 21.2 for three or more connections.[15]

Columns 2-3 of Table 2 report the estimated effects separately for Jews and Arabs, the two main Israeli ethnic groups. Columns 4-5 report the estimates separately for males and females. The estimated impact of weak connections is stronger for Arabs than for Jews: for

[14] To ease visualization, I scale the employment outcome by 100 in all employment specifications. Hence, the results are in terms of percentage points.

[15] The weak-phantom ratio is calculated as follows. The unconditional average probability of working in a non-connected firm (R0) is 0.005 percentage points. Therefore, from equation (1), the estimated (average) probability of working in a firm with one weak connection is $0.005 + 0.052 = 0.057$ percentage points. Likewise, the estimated (average) probability of working in a firm with one phantom connection is $0.005 + 0.012 = 0.017$ percentage points. The ratio between the two probabilities is $0.057/0.017 = 3.28$. The ratios for two and three or more connections are calculated similarly.

workers with one connection, the probability of working in a weak-connected firm is 3.88 times higher than a phantom-connected firm for Arabs and 2.90 times for Jews. This gap widens substantially with additional connections, reaching ratios of 30.5 versus 16.7 for three or more connections. Similarly, the effect of weak connections is stronger for males (ratio of 3.46 for one connection) than for females (2.74), with this pattern persisting across all connection counts.[16]

Overall, the findings here about the positive and large impact of strong connections are consistent with the existing literature (Corak and Piraino 2011; Kramarz and Skans 2014; Staiger 2021). However, the existing literature finds no impact for weak parental connections, such as parents of high-school classmates or high-school classmates of one's parents (Kramarz and Skans 2014; Plug et al. 2018). The positive effect I find for the channel of parents' past coworkers' networks, compared to other channels of indirect parental networks, is consistent with the literature showing the importance of coworker networks for workers' own labor market outcomes (Granovetter 1974; Cingano and Rosolia 2012; Hensvik and Skans 2016; Glitz 2017; Eliason et al. 2023).[17]

3.4 Event-study analysis

To better investigate the timing of the effect, I estimate the time-varying version of equation (1):

$$e_{ij} = \phi_{x(i)j} + \sum_{c=p,w} \sum_{\tau=-5}^{5} \delta^{c,\tau} \cdot D_{ij}^{c,\tau} + \delta^{o} D_{ij}^{o} + \epsilon_{ij} \tag{2}$$

where $D_{ij}^{c,\tau}$ is a dummy variable equal to one if i has one connection of type $c \in \{p, w\}$ (phantom or weak) at firm j, and the last year i's contact worked at firm j was τ years after i's labor-market entry year. D_{ij}^{o} is a dummy variable equal to one if i has any other type of connection at firm j (multiple weak or phantom connections, mixed connections, or strong connections). All other variables are defined as before.[18]

This specification compares the probability of worker i working at a firm where her contact left just before she entered the labor market to the probability of working at a firm

[16] See Section 3.6 below for further discussion of the heterogeneous effects for different groups of workers. See also Appendix C.3 for the robustness of the results to alternative definitions of the labor-market entry year, the ages of the new workers in the sample, and the sample of firms.

[17] See also Section 3.6 below, where I show that the effect of connections decays over time, which might explain why links formed long ago are not useful.

[18] Note that, for $\tau < 0$, the contact left the firm before time zero (the labor-market entry year), therefore $D_{ij}^{w,\tau<0} = 0 \ \forall i, j$. Similarly, if i's contact left the firm at time zero, i cannot have phantom connections at that firm: $D_{ij}^{p,\tau=0} = 0 \ \forall i, j$.

where the contact left shortly after that time. If social connections increase the probability of finding a job at a firm, there should be a discontinuous increase in the estimated effect at time zero. The estimates of the coefficients in equation (2) are plotted in Figure 2—the probability of working at a firm as a function of the lag between the last year the potential contact worked at the firm and the labor-market entry year. Negative lags represent phantom connections, and non-negative lags represent weak connections.[19]

The probability that a new worker began work at a firm that her parental contact left before she entered the labor market was 0.005-0.012 percentage points higher than the probability for another worker with similar observable characteristics but no connections at all. The estimated effect increased to 0.040-0.058 percentage points when the contact left the firm after time zero. The discrete increase in employment probability occurs exactly at time zero—the labor-market entry year, indicating that the presence of the contact at the firm at that time accounts for the change in employment probability.[20]

The estimates from the event study are consistent with the regression results in Table 2. The size of the "jump" in Figure 2 is about 0.04 percentage points, which closely matches the difference between the effects of weak and phantom connections (both with one contact only) reported in Table 2.

Note also that the event-study specification relies on weaker identifying assumptions than the baseline regression in Table 2, since it allows for some trends in the running variable (time since the first job), and the treatment effect is identified by the magnitude of the jump—similar in spirit to a regression discontinuity design. As shown in the previous paragraph, the pre-trends are small relative to the estimated effect, and both the baseline regression and the event-study yield similar estimates.

3.5 Discussion and validation of the identification strategy

My identification strategy assumes that, absent parental connections, there is no systematic difference between the probability of working at a firm with a weak (active) connection and at a firm with a phantom (non-active) connection.

There are three potential threats to this identification strategy:

[19]See Figure A5 for a raw data version of Figure 2. Figure 2 does not show estimates for other connections or phantom connections where the potential contact left the firm after time zero but did not work there at time zero (for example, if she started working at the firm after that time). Table A2 reports all estimated coefficients from equation (2). The estimated effect for other connections is about ten times larger than the estimates for weak connections with one contact. The estimated effects for phantom connections with a positive lag are slightly higher than for other types of phantom connections, but still substantially smaller than the estimates for weak connections.

[20]Similar patterns are obtained when estimating equation (2) for subgroups of workers divided by ethnicity and gender (Figure A6).

1) *Worker-job specific unobservables.* Characteristics unrelated to the impact of connections themselves might make a worker more likely to work at a firm where they have active connections.

2) *Endogenous separations.* The potential contact's departure from the phantom-connected firm before time zero might reflect negative attributes of the relationship between the contact and the firm, which could be correlated with or even transmitted to the worker.

3) *General hiring trend differences.* The contact's departure might reflect a general negative hiring trend in firms that have more phantom connections to workers in the sample compared to firms with more weak connections.

While these threats cannot be entirely ruled out, I present evidence against each:

1) The event-study results suggest that any worker-firm-specific characteristic continuously increasing the hiring probability as a function of the contact's departure timing cannot explain the observed pattern. To challenge this, one would need to propose worker-firm-specific confounds that discretely increase when the potential contact leaves the firm at time zero. Moreover, Appendix C.4 shows that firms with weak and phantom connections are similar in observable worker-firm-specific characteristics, suggesting that they are also similar in unobservable characteristics.

2) To address the possibility of endogenous separations influencing the estimated effects, I estimate the effects using two exogenous causes of separation: coworkers' deaths and retirements. Specifically, I compare the probability of working at firms where parents' coworkers died or retired after the labor-market entry year to firms where contacts died or retired a few years before.[21] These estimates are similar in magnitude to the benchmark result, with odds ratios of 2.6 and 3.9 for the "death" and "retirement" connections, respectively. However, due to much smaller sample sizes, they are less precise and not statistically different from one (Appendix C.5).

3) To check for potential differences in employment trends between firms with weak and phantom connections, I perform a placebo test. This involves assigning a worker's connections to a random worker with similar observable characteristics. I find no hiring differences between phantom and weak connections of a placebo worker (Appendix C.6).

Finally, it is crucial to distinguish between selection to treatment (discussed above) and heterogeneity of effects. The "effect" (the difference between weak and phantom connections) may correlate with observable and unobservable factors. For instance, the effect might be more pronounced when the firm is geographically closer to the worker. This correlation does not invalidate my identification strategy, as I do not assume a constant effect. I extensively examine the heterogeneity of the effect across various observable dimensions in the next

[21] See Azoulay et al. (2010) and Jäger (2016) for early use of death for exogenous variation in networks.

section.

3.6 Heterogeneous effects

To check the heterogeneity of the effect, I re-estimate equation (1) with separate coefficients for different categories of weak and phantom connections. Figure 3 shows the difference between the estimates of the effect of weak and phantom connections on employment for each category. Below are the main findings.[22]

Firm size. Figure 3, Panel A, compares the effect of weak and phantom connections with one contact. The estimates suggest that connections formed in smaller firms are more effective. Moreover, the effect of connections disappears for firms with more than 400 workers. This result is consistent with the intuitive view that the probability or intensity of connections between a random pair of workers is higher in smaller firms, or that parents may not know all former coworkers in large firms. Furthermore, finding a job at a connected firm is more likely when the firm is smaller (Panel B). This can also be explained by a higher probability that the contact can influence, or even be relevant to, hiring decisions in smaller firms.[23]

Duration and time of connection formation. The effect is more substantial the longer the parent and the contact worked together. Likewise, the effect is weaker for connections generated less recently (Figure 3, Panels C and D). Overall, these results are consistent with the findings of Eliason et al. (2023), who show that the impact of social connections is stronger for connections of longer duration, more recently established, or fostered in smaller groups.

Parent's and coworker's salary. I examine the magnitude of the effect as a function of the overall (countrywide) and within-firm salary rank of the parents and coworkers. Starting with the overall salary rank, Panels E and F of Figure 3 show that, except for the two lowest wage percentiles, the effect is smaller the higher the salary of the parents and coworkers, indicating that workers from disadvantaged backgrounds use connections more.[24] In contrast,

[22]Note that when comparing the effects of two groups, one cannot distinguish between the two following alternatives without direct information on the actual connections between the workers: 1) the probability of connection formation is higher for one group compared to another, and 2) the impact of the connections is higher.

[23]Table A3 presents the regression results separately for each of the eight connection types, distinguishing between connections in small firms (fewer than 200 workers) and large firms (200–500 workers) for both past and current firm sizes. The difference between weak and phantom connections is substantially larger in smaller firms across all connection count categories. For instance, for workers with one connection, the difference is 0.065 percentage points for connections generated in small firms compared to only 0.008 percentage points for connections generated in large firms (columns 1–2). For workers with two connections, the differences are 0.442 versus 0.019 percentage points, and for three or more connections, 1.573 versus 0.072 percentage points. This pattern holds for current firm size measures as well.

[24]Either demand or supply factors can explain this pattern. That is, workers from a high socioeconomic background rely less on social connections because they need them less to find a good job or because they

the effect tends to increase with the relative salaries of parents and coworkers within the firm (Panels G and H).[25] Moreover, the smaller the wage gap between the parent and the coworker, the stronger the effect (Panel I).

Gender. The effect is stronger for males than for females. This pattern holds when considering the gender of the worker, the parent, and the parent's coworker (Figure 3, Panels J-L), and is in line with findings in Bayer et al. (2008) and Kramarz and Skans (2014). The mechanism behind this gender difference is not fully understood and remains an open question.[26]

Ethnicity and education. The effect is stronger for Arabs than for Jews and weaker for more highly educated workers (Panels M-O). This is consistent with the results above showing that the effect is stronger for workers from disadvantaged backgrounds.[27]

Similarity between the child, the parent, and the coworker. The effect is stronger if the parent and the worker, or the worker and the parent's coworker, are of the same gender (Panels P-Q). Likewise, the effect is stronger if the worker and the parent's coworker are from the same ethnic group (Panel R).

Unemployment rate. Figure A13 shows the correlation between the estimated effect of weak connections by year and the total unemployment rate in that year. The effect is stronger in years with high unemployment rates, consistent with the results Kramarz and Skans (2014) find for strong parental connections.

3.7 Importance of match quality

Figure 4 shows the heterogeneity of connection effects by three observed measures of the importance of match quality at the firm level: mean match duration, standard deviation of match duration, and wage match effects, which capture how much the specific worker-firm pairing matters for wages beyond worker and firm characteristics alone (measured as the firm average absolute residual in the AKM regression, estimated without the new workers'

(or their parents) prefer that they find a job without the help of social connections. I discuss this pattern further in Section 5.1, where I argue that it is consistent with connections being relatively more valuable in segments of the labor market where formal screening returns less.

[25]The fact that links to higher-ranking employees within a firm are more effective is consistent with Montgomery (1991)'s model. See Kramarz and Skans (2014) for similar results.

[26]The gender gap persists when controlling for sector. Figure A12 shows the effects by gender separately for each of the main sectors (Mining and Other sectors are excluded from the figure due to small sample sizes). The point estimate for males is larger than that for females in all sectors except the public sector. The average gap between males and females across sectors (0.028, s.e. 0.013) is very close to the overall gender effect without controlling for sectors (0.033), suggesting that the gender difference in the effectiveness of parental connections is not driven by differential sorting into sectors.

[27]Kramarz and Skans (2014) also find that the effect of parental ties is stronger for young workers with less education, lower GPA grades, and generally weaker labor market prospects.

sample). Firms with lower values on these measures—shorter expected tenure, less variation in tenure, and smaller wage match effects—are settings where the quality of the worker-firm match is likely less important. In such firms, employment may be more temporary or standardized, and wages may depend more on observable worker and firm attributes than on the idiosyncratic fit between a particular worker and firm. The left column shows the effects across deciles of each characteristic, while the right column presents the sector-level correlations between these characteristics and the connection effect.

The results suggest that connections are indeed more effective in settings where match quality is less crucial. Panels A and B show that the connection effect is stronger in firms with lower average match duration. While the estimates in the first two bins of Panel A are noisy, excluding these bins reveals a negative pattern, which is strongly supported by the sector-level analysis in Panel B. Panels C and D indicate that the effect is more pronounced in firms with lower variance in match duration, with both the bin analysis and sector correlations showing a clear negative relationship. Finally, Panels E and F demonstrate that firms with smaller match effects—where the quality of the worker-firm pairing matters less for wages—exhibit larger connection effects. For this measure, the bin analysis in Panel E shows a clear negative pattern, while the sector-level correlation in Panel F is less pronounced (although the slope remains negative).

This heterogeneity suggests that parental connections primarily work through alleviating information frictions rather than improving match quality. If connections mainly helped workers find better matches, we would expect the effect to be stronger in environments where match quality is more important—that is, in firms with higher match duration, greater variance in duration, and larger match effects. Instead, the opposite pattern emerges: connections matter most precisely in settings where matches are more homogeneous and less consequential. This suggests that the primary mechanism is reducing search costs and facilitating job finding, rather than identifying superior match-specific opportunities. This conclusion is also supported by the modest effect of connections on wages and duration at the job, which I find in Section 4.

3.8 Employment and job switch rates

The above estimates demonstrate sizable effects of parental connections on matching rates. An important related question is whether the overall stock of connections also affects transitions between employment states, from non-employment to employment and between employers. The main identification strategy used throughout this paper, comparing employment at firms with weak versus phantom connections, cannot be used to assess the effect of the overall stock of connections on overall employment or transition probability, as the definition of weak versus phantom connections is worker-firm specific. The estimates in this section should therefore be interpreted with caution as they reflect correlations rather than causal effects, since workers' network characteristics may be correlated with unobserved factors that also affect employment and job mobility.

To provide suggestive evidence on this question, I construct two measures of overall connections: employment share (the fraction of contacts currently employed at time t, following Cingano and Rosolia (2012)) and log number of contacts. I then estimate panel regressions where the outcomes are (1) whether the child is employed at year t, and (2) whether the child switched jobs between years t and $t+1$. The sample includes individuals born between 1979 and 1993 who are observed working in a firm with 5-500 employees at least once between 2006 and 2015 at ages 22 to 27. I track these workers annually during this age range, recording their employment status and firm affiliation across all firms.

Table 3 presents the results. A one-standard-deviation increase in employment share is associated with an employment probability increase of 0.09 percentage points (column 1), while a one-standard-deviation increase in log contacts is associated with an increase of 0.17 percentage points (column 2). Column 3, which includes both measures jointly, shows similar effects.

The job switching results (columns 4 to 6) show a contrasting pattern for the two measures of connections. Column 4 shows that a higher employment share of connections is associated with lower switching probability (a decrease of 0.24 percentage points), consistent with workers staying longer at firms where they have more connections. However, the log contacts specification (column 5) shows a positive correlation with switching probability (an increase of 0.86 percentage points), suggesting that workers with larger networks switch jobs more frequently overall. Column 6, which includes both measures jointly, confirms these patterns.

In summary, the correlational evidence suggests that both the quality and quantity of parental connections are associated with labor market outcomes, though in different ways. Workers whose connections have higher employment rates (higher quality networks) are more likely to be employed and less likely to switch jobs, consistent with more stable employment relationships facilitated by active networks. In contrast, workers with more total connections (larger networks) are more likely to be employed and also more likely to switch jobs, suggesting that larger networks facilitate both entry into employment and mobility between employers.

4 Salary and job tenure

4.1 Regression framework

Next, I examine the relationships between parental links and other labor market outcomes for new workers. Specifically, I compare the wage, tenure at the first job, and salary growth after three years for observably similar workers within the same firm, distinguishing between those with and without connections to the firm. To account for factors correlated with parental connections, I compare weak and phantom connections.[28]

Precisely, I assume that the labor market outcome y_i of a new worker i equals:

$$y_i = \sum_c \delta^c D^c_{i,j(i)} + \phi_{x(i)} + \psi_{j(i)} + \epsilon_i, \tag{3}$$

where $D^c_{i,j(i)}$ is an indicator variable capturing whether a worker i has connections of type c at her first job, where the possible types of connections are phantom (1,2, and 3+ connections), weak (similarly), mixed, and strong. $\phi_{x(i)}$ and $\psi_{j(i)}$ are group and firm fixed effects, respectively. As before, the workers' groups include all combinations of ethnicity, gender, education, age, year of the first job, and district of residence of the new workers.

4.2 Regression results

Column 1 of Table 4 reports the estimates of equation (3) with log salary as the outcome variable. Within the same firm, the salary of workers with one phantom connection is 1.1 percent higher than that of observably similar workers without connections. Workers with one weak connection earn a premium of 2.6 percent over similar workers without connections. Compared to those with phantom connections, workers with one weak connection earn 1.5 percent more. This difference increases with the number of connections: the gap between weak and phantom connections is 3.1 percent for two connections and 5.6 percent for three or more connections. Workers with mixed connections earn 3.6 percent more, and workers with strong connections earn 12.0 percent more than similar workers at the same firm without connections.

Column 2 investigates whether workers with connections at their first firm stay there for longer periods than unconnected workers. The outcome variable is the number of years the worker stayed at her first firm. The first-job duration of workers with one phantom connection is 0.093 years higher than that of workers without connections, while workers with one weak

[28] While comparing hiring with weak vs phantom connections addresses many of the endogeneity concerns, the identification argument in this section is weaker than the firm-specific employment effects presented in the previous section, as connections might affect the probability of working in some firms and not others.

connection stay 0.184 years longer. Compared to phantom connections, workers with one weak connection stay 0.091 years longer. For two connections, the difference between weak and phantom is 0.243 years, and for three or more connections, the gap is 0.118 years. Workers with mixed connections stay 0.256 years longer, and workers with strong connections stay 0.589 years longer than unconnected workers.

Column 3 compares salary growth after three years (not necessarily at the same job) for workers who started their first job at a connected versus an unconnected firm. The salary growth of workers who started their first job at a firm with one weak connection is 2.7 percentage points lower than that of those with one phantom connection. For two connections, the difference is 5.5 percentage points, and for three or more connections, the gap is 4.7 percentage points. Workers with mixed connections experience 0.8 percentage points lower salary growth, and workers with strong connections experience 1.8 percentage points higher salary growth compared to workers without connections.

The associations between connections and salary, tenure, and wage growth vary by ethnicity and gender. Appendix Tables A7 and A8 present results separately by ethnicity (Jews and Arabs) and gender (males and females). For log salary, the weak-phantom difference at one contact is similar across ethnicities (1.5 percent for Jews and 1.8 percent for Arabs) but widens with the number of contacts, reaching 8.9 percent for Jews and 2.4 percent for Arabs at three or more contacts. By gender, the pattern at one contact is a 2.0 percent premium for males and essentially zero for females, but for two and three or more contacts the difference between males and females reverses. For job tenure, weak connections raise tenure for both ethnic groups, with substantially larger effects for Jews than Arabs at multiple contacts. Salary growth differences are negative across most groups, with no clear pattern for the differences between ethnic and gender groups.

4.3 Event-study analysis

Next, I estimate event-study specifications analogous to equation (2) for log salary:

$$y_i = \phi_{x(i)} + \psi_{j(i)} + \sum_{c=p,w} \sum_{\tau=-5}^{5} \delta^{c,\tau} \cdot D^{c,\tau}_{i,j(i)} + \delta^o D^o_{i,j(i)} + \epsilon_i \tag{4}$$

where y_i is the log salary for worker i at their first job $j(i)$. $D^{c,\tau}_{i,j(i)}$ is a dummy variable equal to one if worker i has one connection of type $c \in \{p, w\}$ (phantom or weak) at firm $j(i)$, and the last year the contact worked at firm $j(i)$ was τ years relative to i's labor-market entry year. $D^o_{i,j(i)}$ captures other connection types. $\phi_{x(i)}$ and $\psi_{j(i)}$ represent group and firm fixed effects, respectively.

Figure 5 presents the results for workers with exactly one contact, comparing outcomes at firms where the contact left before versus after the worker's labor-market entry year. Although results are noisy, the figure indicates that periods after time zero are associated with higher salaries compared to those before. Prior to time zero, there is no clear pattern in salary differences between workers at firms with phantom connections and those without connections. After time zero, workers with weak connections earn a premium of up to 5.9 percent over similar workers with no connections, with an average association of 2.9 percent.

4.4 Dynamics

To examine how the associations with connections evolve over time, I estimate a dynamic version of equation (3):

$$y_{it} = \phi_{x(i)} + \psi_{j(i)} + \sum_{c} \sum_{l=0}^{4} \delta^{c,l} \cdot D^{c}_{i,j(i)} \cdot \mathbf{1}\{t = l\} + \beta X_{it} + \epsilon_{it} \tag{5}$$

where y_{it} is the outcome variable (log salary or retention) for worker i at time t years after their labor-market entry year. $D^{c}_{i,j(i)}$ indicates whether worker i has connections of type c at their first job $j(i)$. $\mathbf{1}\{t = l\}$ is an indicator function equal to one when the observation period t equals l. $\phi_{x(i)}$ represents group fixed effects, constructed using all combinations of workers' observable characteristics (ethnicity, education, gender, year of first job, age, and district of residence), and $\psi_{j(i)}$ represents first-firm fixed effects. X_{it} includes calendar year, period (years since entry), and age fixed effects. The specification estimates separate coefficients for weak and phantom connections (with one contact), as well as other connection types, for each year since labor market entry ($l = 0$ to 4). Note that unlike the event-study specifications in equations (2) and (4), the running variable here is time since the worker's labor market entry (not the last year the contact worked at the firm).

Figure 6 presents the differences between the coefficients of weak and phantom connections with one contact as a function of years since labor-market entry. For log salary, the difference between weak and phantom connections is positive and significant at entry (2.1 percent), then smaller and not statistically distinguishable from zero in subsequent years (Panel A). Regarding retention (an indicator for staying at the firm in the following year), the association between weak and phantom connections is positive at year 0 (5.3 percent), indicating a higher probability of staying at the firm for workers with weak connections relative to those with phantom connections. This gap diminishes over time and reverses by year 2.

A natural extension of this analysis is to examine whether the advantages from parental connections translate into greater cumulative lifetime earnings and higher earnings in subsequent jobs. To address this, I estimate equation (3) with two alternative outcomes: the discounted sum of monthly salaries over the first five years in the labor market (discounted at a 4 percent annual rate), and the average yearly salary in the first year of the second job for workers who transition to a second employer within five years of labor market entry. As before, I compare weak and phantom connections separately for each connection count (1, 2, and 3+ connections).

Table 5 presents the results. Column 1 examines the correlation between connections and discounted cumulative salaries over the first five years in the labor market, controlling for first-firm fixed effects. The discounted salary of workers with one phantom connection is higher by 3.4 percent compared to workers without connections, while workers with one weak connection earn 4.2 percent more. Compared to phantom connections, workers with one weak connection earn essentially the same amount. For two connections, the difference between weak and phantom is 3.9 percent and is not statistically significant. For three or more connections, the difference is 6.2 percent. Workers with mixed connections earn 7.6 percent more, and workers with strong connections earn 21.3 percent more than unconnected workers.

Column 2 examines the correlation between connections at the first job and average yearly salary in the first year of the second job, with first-firm fixed effects. Most correlations are small and statistically insignificant. The differences between weak and phantom connections are also not significant and, if anything, somewhat negative: 0.6 percent lower for one connection, 0.5 percent higher for two connections, and 5.3 percent lower for three or more connections.

In summary, the earnings advantages associated with weak parental connections are modest and primarily concentrated among workers with multiple connections. For single connections, the differences in cumulative five-year earnings between weak and phantom connections are essentially zero, rising to an imprecisely estimated 6 percent only for workers with three or more connections. More strikingly, weak connections show no distinct advantage in subsequent job salaries, with differences that are small, statistically insignificant, and, if anything, slightly negative. Combined with the findings that initial wage premiums and retention patterns dissipate over time, the evidence suggests that any advantages are confined to the initial employment relationship, with no observed spillover to subsequent employers.

5 Discussion

The evidence in this paper suggests that weak parental ties facilitate labor market entry primarily by transmitting information about firm vacancies, rather than by improving the worker-firm match. This section lays out the case for this mechanism and compares my findings to Kramarz and Skans (2014).

5.1 Mechanisms

The interpretation that connections operate primarily through an information channel about vacancies is supported by the heterogeneity analysis in Section 3.7. If connections primarily improved match quality by helping workers find firms where they are particularly productive, effects should be stronger in environments where match quality matters more, such as firms with higher average tenure, greater variance in tenure, and larger match effects on wages. Instead, the opposite pattern emerges: connection effects are strongest precisely where matches are more homogeneous and less consequential for outcomes. This finding is difficult to reconcile with a match-optimization story but follows naturally if connections primarily help workers learn about and gain access to job openings they would otherwise miss.

Additional evidence reinforces this interpretation. First, the wage premium is modest in magnitude, at 1.5 to 5.6 percent, and declines over time. Second, weak connections provide no advantage in subsequent job salaries after workers leave their first employer. If connections improved match quality in ways that enhanced worker productivity or revealed valuable worker characteristics, these advantages should translate into higher wages at future employers who observe the worker's performance. The absence of such spillovers suggests that any benefits are firm-specific and transitory, consistent with connections helping workers overcome information barriers at a particular firm rather than identifying fundamentally better matches. Third, the finding that effects are larger for disadvantaged groups aligns with a search-friction interpretation. At the low-skill end of the labor market, the returns to careful screening for unobserved match quality are smaller, so the marginal value of information about job openings from connections is correspondingly higher.

A specific variant of the match-quality channel deserves separate consideration. In Jovanovic (1979)'s framework, when new entrants' productivity is noisy from the firm's perspective, the firm offers a low initial wage and rapid wage growth, with high turnover as low-productivity workers exit. If connections reduce this uncertainty, the model predicts higher initial wages, slower wage growth, and longer tenure for connected workers (Dustmann et al. 2016). The wage and career patterns documented here are consistent with this variant. However, two pieces of evidence indicate that any uncertainty-reduction channel is at most a secondary mechanism alongside the primary information channel about vacancies. First, the magnitude of the wage and tenure effects is small relative to the large employment effects of weak connections. Second, the employment effects are largest in firms where match quality matters least, which a worker-quality-revelation story cannot easily explain.

A specific mechanism consistent with the full pattern of evidence is one in which insiders hear of vacancies within the firm first and may pass that information to people they know. Under this mechanism, the entry-wage premium for connected workers reflects the firm's willingness to pay to fill a recently vacated position quickly, rather than its belief that the worker is of higher quality. This predicts exactly the pattern observed in the data: a wage premium concentrated at entry, no productivity-related wage growth, and no portable returns at subsequent employers.

Taken together, the evidence points consistently toward connections operating primarily through the information channel emphasized by Granovetter (1973) and a long line of theoretical models of job search: weak ties are valuable for "knowing about appropriate job openings at just the right time" rather than for systematically improving the value of the worker-firm match.

5.2 Comparison with Kramarz and Skans (2014)

The most direct point of comparison for these results is Kramarz and Skans (2014) (KS hereafter), who study parental connections and labor-market entry using Swedish administrative data. My estimates of strong (parental) connection effects are quantitatively comparable to theirs. I find that workers with a parent at a firm are 3.1 percentage points more likely to find their first job there than observably similar workers without such a parent; KS find a corresponding effect of 6.1 percentage points. Both papers find positive wage growth in the first few years, with my estimates smaller in magnitude: 1.8 percent over three years here, and 5.4–7.3 percent in KS. On tenure, strong-tie hires stay at their first firm substantially longer in both papers, although the outcome measures are not directly comparable: I find that strong-tie hires stay 0.59 years longer at their first firm, while KS find that they are 5.7 to 12 percentage points more likely to remain at the same firm after three years.[29] Both papers also find that strong-tie effects are larger for more disadvantaged workers—low-educated graduates with low grades in KS; Arabs, less-educated workers, and those from lower socioeconomic backgrounds in this paper. Finally, both papers also find larger effects

[29] The two papers do diverge on entry wages: KS find that strong-tie hires earn lower wages than other graduates entering the same plant, while I find that strong-tie hires earn higher wages than observably similar workers at the same firm.

for males than for females, in line with Bayer et al. (2008).

The papers diverge on weak ties: KS find that classmates' parents have negligible effects on hiring, while I find substantial effects of parents' past coworkers. The difference probably reflects the construction of the weak tie rather than a contradiction in findings: the chain from worker to classmate to classmate's parent contains no professional link at any step; by contrast, a parent's past coworker has a professional relationship with the parent and is currently inside a specific firm whose openings they may share. Read together, the two papers suggest that whether a "weak tie" matters depends on how the tie is constructed—the presence of a professional relationship appears to matter, with such ties showing large positive effects while ties without them show negligible effects.

6 Conclusion

This paper studies how parents' professional networks shape the labor-market entry of young workers. The empirical strategy compares firms where a parent's past coworker was present at the worker's labor-market entry year to firms where the coworker had left shortly before or arrived shortly after.

I find that workers are 3.3 times more likely to find their first job at a firm with an active connection than at a firm with a phantom one, with effects rising substantially in the number of connections. Effects are stronger when the parent and coworker overlapped recently and for longer durations, when the contact was at a smaller firm, and when the worker, parent, and coworker share characteristics. Connection effects are also larger for workers from disadvantaged backgrounds—Arabs, less-educated workers, and those from lower socioeconomic backgrounds—suggesting that informal networks are relatively more valuable in segments of the labor market where formal channels are weaker.

Connected workers earn modestly higher entry wages and stay at their first firm somewhat longer, but they experience slower wage growth and show no advantage in subsequent jobs. Combined with the finding that connection effects are strongest in firms where match quality matters least, this pattern is consistent with connections primarily transmitting information about specific firm vacancies rather than improving the worker-firm match. The evidence is also consistent with the assumption, common in theoretical models of job search, that different labor-market participants learn about new vacancies at different rates.

The findings have nuanced implications for policy. If weak ties primarily reduce search frictions, then public policy can be designed to replicate their effects without relying on personal networks. Interventions such as encouraging job interviews or subsidizing internships at high-quality firms for disadvantaged workers could prove effective, since these workers typically have fewer and lower-quality connections. Given that connections formed in smaller firms, more recently, and among individuals with shared characteristics are most potent, such interventions should prioritize intensive, recent, and relatable interactions. However, the findings also suggest that policies strictly prohibiting the use of connections might inadvertently increase inequality, since workers from disadvantaged backgrounds rely most heavily on social links to navigate the labor market.[30]

Finally, this paper documents a pattern that it does not fully explain: connection effects are larger for males than for females. This pattern is consistent with findings in earlier work (Bayer et al. 2008; Kramarz and Skans 2014), but the mechanism behind it remains an open question. Further research is needed to understand the sources of this gender difference and its implications for the role of social networks in labor-market inequality.

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[30] One example is the "Rooney Rule," which requires NFL teams to interview at least one minority candidate for head coaching positions. Solow et al. (2011) find no evidence that this rule has increased the number of minority coaches, likely because search frictions are negligible in such a small, transparent market. Similar mandates in large-scale labor markets, where information barriers are more significant, may yield different results.

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Tables and figures

Table 1: Summary statistics, new workers

	All	Ethnicity		Gender	
		Jews	Arabs	Males	Females
N.	220,806	157,023	63,783	126,233	94,573
Arabs	0.289	0.000	1.000	0.365	0.187
Females	0.428	0.490	0.277	0.000	1.000
College	0.226	0.251	0.164	0.150	0.328
First job					
Age	24.00	24.22	23.48	23.82	24.25
Salary	5,839	6,053	5,312	6,223	5,325
Firm rank	0.604	0.636	0.524	0.597	0.613
Connections					
Weak: 1	0.027	0.021	0.042	0.032	0.021
Weak: 2	0.005	0.003	0.009	0.006	0.003
Weak: 3+	0.004	0.003	0.008	0.005	0.002
Strong	0.071	0.058	0.102	0.086	0.050
Connections					
Av. firm rank					
Weak: 1	0.636	0.660	0.577	0.630	0.646
Weak: 2	0.652	0.675	0.582	0.645	0.662
Weak: 3+	0.668	0.692	0.589	0.659	0.679
Strong	0.623	0.650	0.554	0.614	0.635
N. firms					
Weak: 1	42.65	49.48	25.84	40.67	45.29
Weak: 2	1.858	2.099	1.267	1.783	1.960
Weak: 3+	0.489	0.539	0.365	0.472	0.511
Strong	2.139	2.228	1.921	2.121	2.164

Notes: This table reports summary statistics for the sample of new workers. The first column reports the average value of the variables for the entire sample, and the other columns report for sub-samples separated according to ethnicity and gender. Firm rank is the rank of the firm-specific pay premium estimated using an AKM model (Abowd et al. 1999). "Connections" indicates the share of workers who have weak or strong connections at the first job, where weak connections are defined as firms where the worker's parents' past coworkers were employed during the worker's entry year, and strong connections are firms where the worker's parents themselves worked. The numbers 1, 2, and 3+ indicate the number of such connections at the firm. Av. firm rank of connections is the average firm rank of firms with which the worker has weak or strong connections. N. firms is the number of firms with which the worker has weak or strong connections.

Table 2: Effects of parental connections on firm assignment

	All	Jews	Arabs	Males	Females
	(1)	(2)	(3)	(4)	(5)
Phantom: 1	0.012	0.008	0.032	0.014	0.008
	(0.001)	(0.001)	(0.002)	(0.001)	(0.001)
Phantom: 2	0.039	0.029	0.075	0.042	0.032
	(0.004)	(0.004)	(0.012)	(0.004)	(0.009)
Phantom: 3+	0.045	0.034	0.078	0.045	0.043
	(0.008)	(0.007)	(0.019)	(0.009)	(0.015)
Weak: 1	0.052	0.032	0.143	0.059	0.033
	(0.002)	(0.002)	(0.007)	(0.002)	(0.004)
Weak: 2	0.302	0.172	0.798	0.348	0.187
	(0.023)	(0.019)	(0.084)	(0.027)	(0.042)
Weak: 3+	1.025	0.628	2.397	1.180	0.644
	(0.085)	(0.072)	(0.260)	(0.107)	(0.126)
Mixed (phantom/weak)	0.248	0.161	0.604	0.261	0.215
	(0.009)	(0.008)	(0.034)	(0.010)	(0.019)
Strong	3.091	2.597	4.497	3.105	3.050
	(0.059)	(0.065)	(0.137)	(0.071)	(0.122)
Diff. weak-phantom: 1	0.040	0.024	0.111	0.045	0.025
	(0.002)	(0.002)	(0.007)	(0.002)	(0.004)
Diff. weak-phantom: 2	0.263	0.143	0.723	0.306	0.155
	(0.024)	(0.020)	(0.086)	(0.027)	(0.044)
Diff. weak-phantom: 3+	0.981	0.594	2.32	1.135	0.601
	(0.086)	(0.073)	(0.263)	(0.108)	(0.125)
Ratio weak-phantom: 1	3.280	2.897	3.883	3.457	2.741
	(0.168)	(0.184)	(0.278)	(0.194)	(0.308)
Ratio weak-phantom: 2	6.962	5.256	10.15	7.641	5.415
	(0.819)	(0.911)	(1.885)	(0.918)	(1.880)
Ratio weak-phantom: 3+	21.23	16.71	30.46	24.61	14.61
	(3.873)	(3.688)	(8.902)	(5.328)	(6.065)
R0 (no connections)	0.005	0.005	0.006	0.005	0.006
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Observations	21,166,443	16,837,526	4,328,917	15,319,313	5,847,130

Notes: This table reports the mean (and standard deviation) of the estimated coefficients of phantom, weak, mixed, and strong connections across 100 estimations of equation (1) using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points. Phantom and weak connections are categorized by the number of contacts: one, two, or three or more. "Diff. weak-phantom" reports the difference between the coefficients of weak and phantom connections for each category. "Ratio weak-phantom" is the estimated odds ratio between working at a firm with weak connections and working at a firm with phantom connections. The first column reports results for the entire sample, while columns 2-3 report results by ethnicity (Jews and Arabs) and columns 4-5 report results by gender (Males and Females).

Table 3: Effect of connections on employment and job switching

	Employment			Switch		
	(1)	(2)	(3)	(4)	(5)	(6)
Employment share	0.091		0.089	-0.237		-0.251
	(0.023)		(0.023)	(0.036)		(0.036)
Log contacts		0.170	0.169		0.858	0.862
		(0.023)	(0.023)		(0.039)	(0.039)
Observations	1,815,344	1,815,344	1,815,344	1,815,344	1,815,344	1,815,344

Notes: This table reports the correlation between measures of quality and quantity of weak parental connections on the one hand and child employment and job switching on the other. The outcome in columns 1–3 is whether the child is employed at year t. The outcome in columns 4–6 is whether the child switched jobs between years t and $t + 1$. Employment share measures the fraction of weak connections currently employed at a firm with 5–500 workers at time t. Log contacts is the log of one plus the number of connections. The outcomes are scaled by 100, and the covariates are divided by their standard deviation, so results are in percentage points for an increase of one standard deviation. All specifications include group fixed effects. Robust standard errors clustered by group are reported in parentheses.

Table 4: Correlation between parental connections at first job and log salary, job tenure, and salary growth

	Log salary	Job tenure	Salary growth
	(1)	(2)	(3)
Phantom: 1	0.011	0.093	0.008
	(0.004)	(0.021)	(0.010)
Phantom: 2	0.015	0.101	0.051
	(0.008)	(0.049)	(0.024)
Phantom: 3+	0.012	0.219	0.037
	(0.011)	(0.063)	(0.029)
Weak: 1	0.026	0.184	-0.020
	(0.004)	(0.022)	(0.012)
Weak: 2	0.045	0.344	-0.004
	(0.011)	(0.061)	(0.026)
Weak: 3+	0.068	0.337	-0.011
	(0.013)	(0.064)	(0.042)
Mixed (phantom/weak)	0.036	0.256	-0.008
	(0.004)	(0.025)	(0.010)
Strong	0.120	0.589	0.018
	(0.004)	(0.026)	(0.009)
Diff. weak-phantom: 1	0.015	0.091	-0.027
	(0.005)	(0.030)	(0.016)
Diff. weak-phantom: 2	0.031	0.243	-0.055
	(0.013)	(0.079)	(0.036)
Diff. weak-phantom: 3+	0.056	0.118	-0.047
	(0.017)	(0.088)	(0.051)
Mean (no connections)	8.577	1.933	0.313
Observations	220,806	220,806	106,368

Notes: This table reports the correlation between parental connections at the first job and log salary, job tenure, and salary growth. The outcome in column 1 is log monthly salary in the first year of the first job. The outcome in column 2 is the number of consecutive years the worker remained at their first job (truncated in 2015). The outcome in column 3 is wage growth after three years in the labor market (only for the 2006–2012 cohorts). Phantom and weak connections are categorized by the number of contacts: one, two, or three or more. "Diff. weak-phantom" reports the difference between the coefficients of weak and phantom connections for each category. All columns include group and firm fixed effects. Groups are constructed using all combinations of workers' observable characteristics (ethnicity, education, gender, year of first job, age, and district of residence). Robust standard errors clustered at the firm level are reported in parentheses.

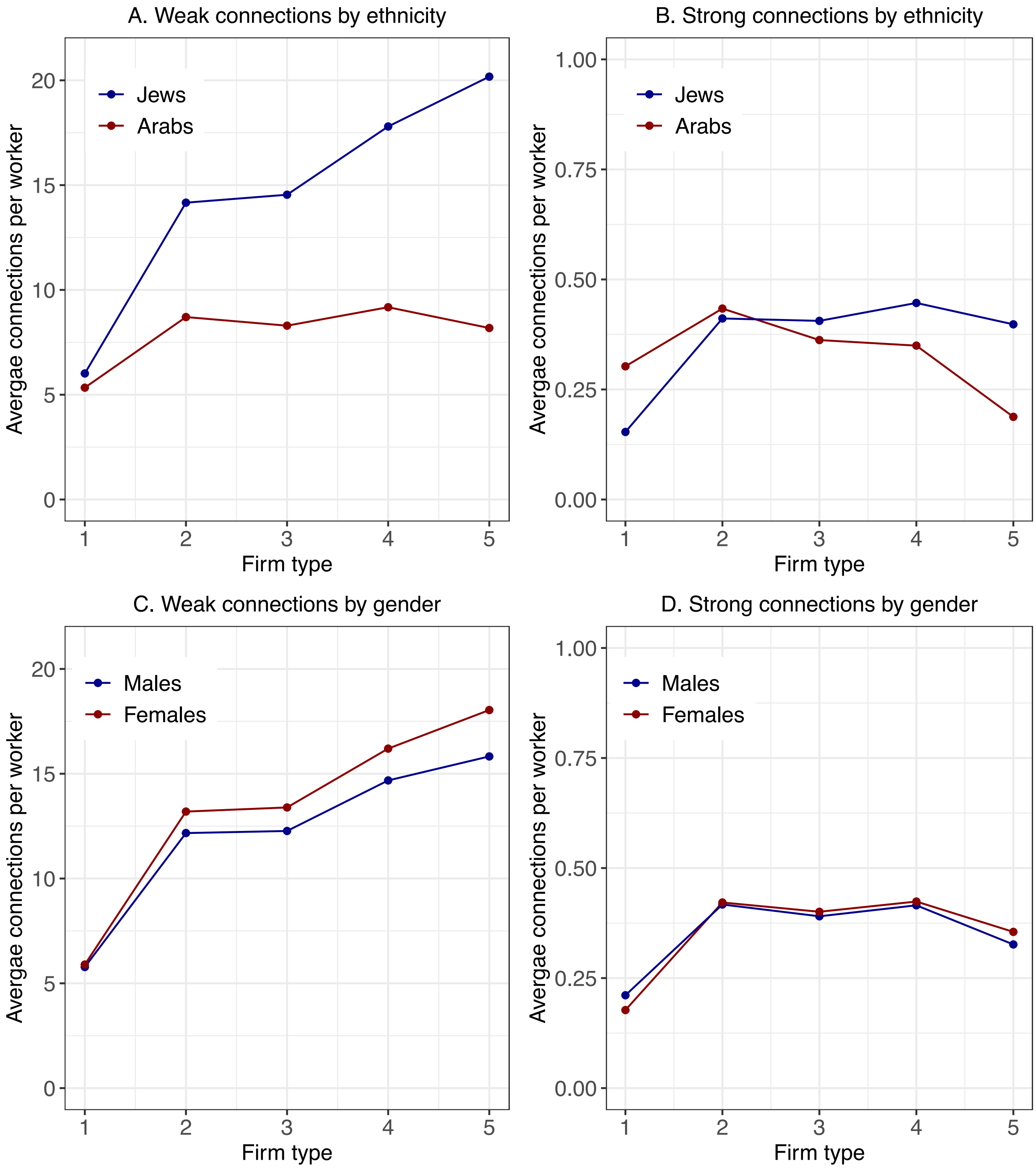
Table 5: Correlation between parental connections at first job and discounted salary and subsequent salary

	Log discounted salary	Log salary (subsequent)
	(1)	(2)
Phantom: 1	0.034	0.021
	(0.008)	(0.017)
Phantom: 2	0.045	0.069
	(0.017)	(0.038)
Phantom: 3+	0.073	0.079
	(0.023)	(0.049)
Weak: 1	0.042	0.016
	(0.008)	(0.019)
Weak: 2	0.084	0.075
	(0.020)	(0.049)
Weak: 3+	0.135	0.026
	(0.026)	(0.061)
Mixed (phantom/weak)	0.076	0.044
	(0.008)	(0.020)
Strong	0.213	0.062
	(0.006)	(0.016)
Diff. weak-phantom: 1	0.007	-0.006
	(0.011)	(0.025)
Diff. weak-phantom: 2	0.039	0.005
	(0.027)	(0.063)
Diff. weak-phantom: 3+	0.062	-0.053
	(0.033)	(0.076)
Mean (no connections)	10.733	10.359
Observations	220,806	134,370

Notes: This table reports the correlation between parental connections at the first job on the one hand and discounted salary and subsequent yearly salary on the other hand. Column 1 reports estimates of equation (3) where the outcome is the log of cumulative discounted salary over the first five years after labor-market entry (annual discount rate of 4%). Column 2 uses as outcome the log of the average yearly salary at the worker's second employer, in the first year at that employer, restricted to workers who change employers within five years of labor-market entry. Both specifications include first-firm fixed effects and group fixed effects, where groups are defined by all combinations of ethnicity, gender, education, age, year of first job, and district of residence. Phantom and weak connections are reported separately by number of contacts (1, 2, 3+). The lower panel reports the implied weak-minus-phantom difference within each contact count. Standard errors in parentheses, clustered at the firm level.

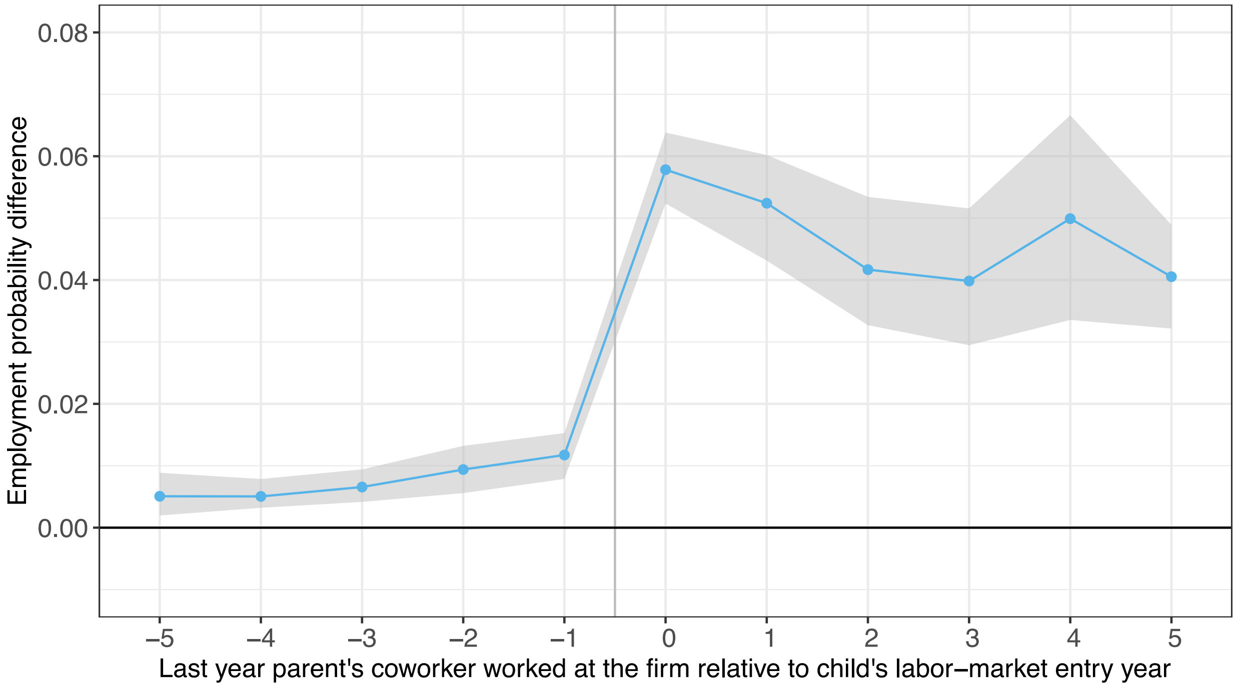
Each column reports OLS coefficients from a separate regression. Column 1 reports estimates of equation (3) where the outcome is the log of cumulative discounted salary over the first five years after labor-market entry (annual discount rate of 4%). Column 2 uses as outcome the log of the average yearly salary at the worker's second employer, in the first year at that employer, restricted to workers who change employers within five years of labor-market entry. Both specifications include first-firm fixed effects and group fixed effects, where groups are defined by all combinations of ethnicity, gender, education, age, year of first job, and district of residence. Phantom and weak connections are reported separately by number of contacts (1, 2, 3+). The lower panel reports the implied weak-minus-phantom difference within each contact count. Standard errors in parentheses, clustered at the firm level.

Figure 1: Average connected firms per worker by worker characteristics, firm type, and connection type



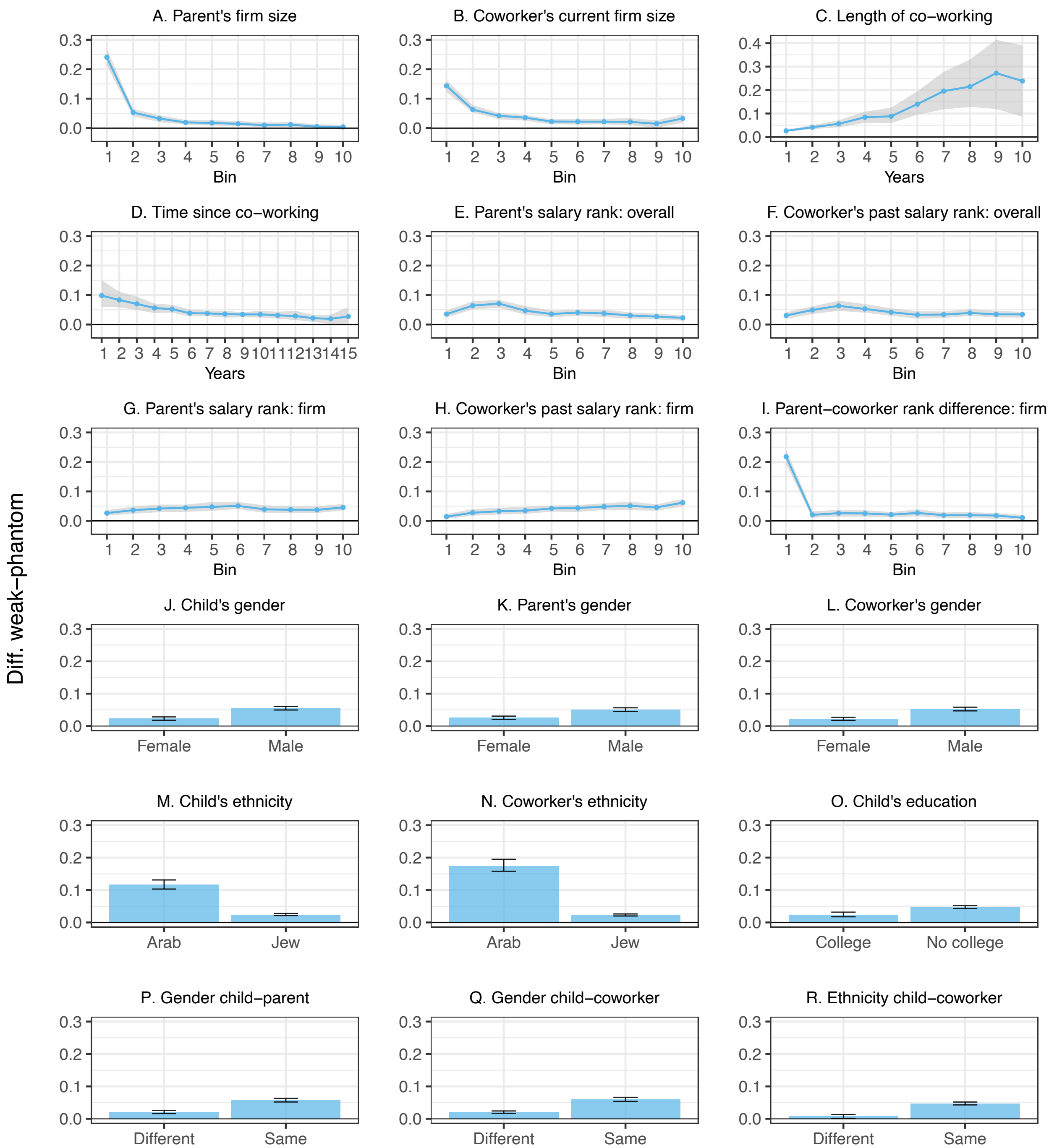
Notes: This figure shows the average number of weakly and strongly connected firms per worker by workers' ethnicity and gender, and by quintiles of the AKM firm premium, averaged over the years 2006-2015. Weak connections are defined as firms where the worker's parents' past coworkers were employed during the worker's entry year. Strong connections are firms where the worker's parents themselves worked. The figure shows weak connections with exactly one contact.

Figure 2: Event-study plot of coefficients: Effect of weak parental connections on firm assignment



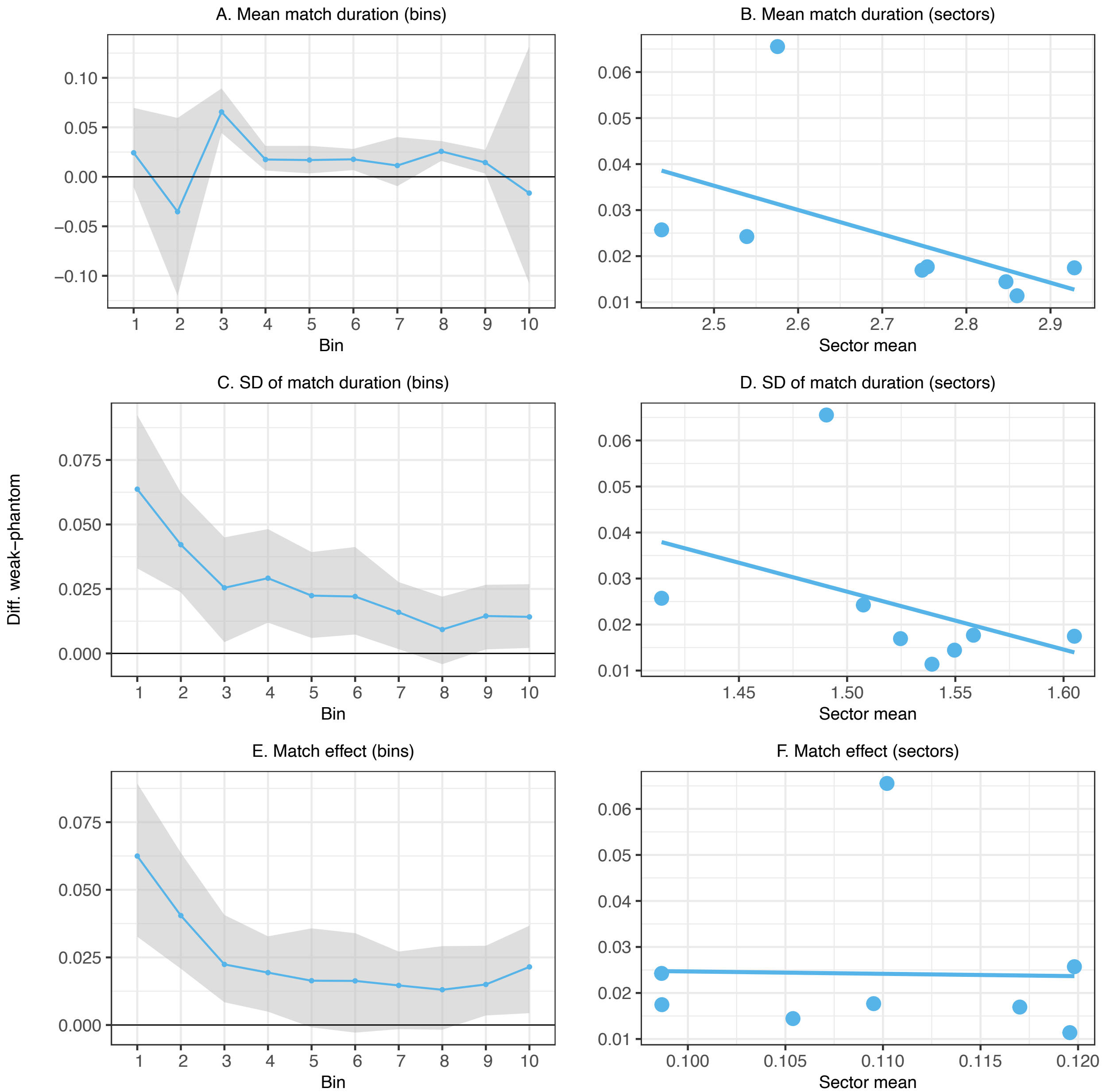
Notes: This figure shows the probability of working in a firm as a function of the difference between the last year the parent's coworker worked at the firm and the worker's labor-market entry year relative to working in a non-connected firm. The figure shows the mean (and 95 percent confidence interval) of the estimated coefficients of phantom and weak connections with one contact across 100 estimations of equation (2) using a 20 percent random sample of workers each time. The employment outcome is scaled by 100. The vertical line between -1 and 0 indicates the change from worker-firm pairs with phantom connections to pairs with weak connections. Table A2 reports all estimated coefficients from equation (2), including other connection types.

Figure 3: Effects of weak parental connections on firm assignment: Heterogeneity by characteristics of the workers and the connections



Notes: Each figure shows the probability of working in a firm with weak connections (with one contact) for different characteristics of the workers and the connections, relative to the probability of working in a phantom-connected firm (with one contact). The points are the mean differences between the coefficients of weak and phantom connections across 100 estimations of equation (1), using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points. The bounds of the 95 percent confidence intervals are constructed using the 2.5 and 97.5 percentiles of the distribution of the differences.

Figure 4: Effects of weak parental connections on firm assignment: Heterogeneity by firm and match characteristics

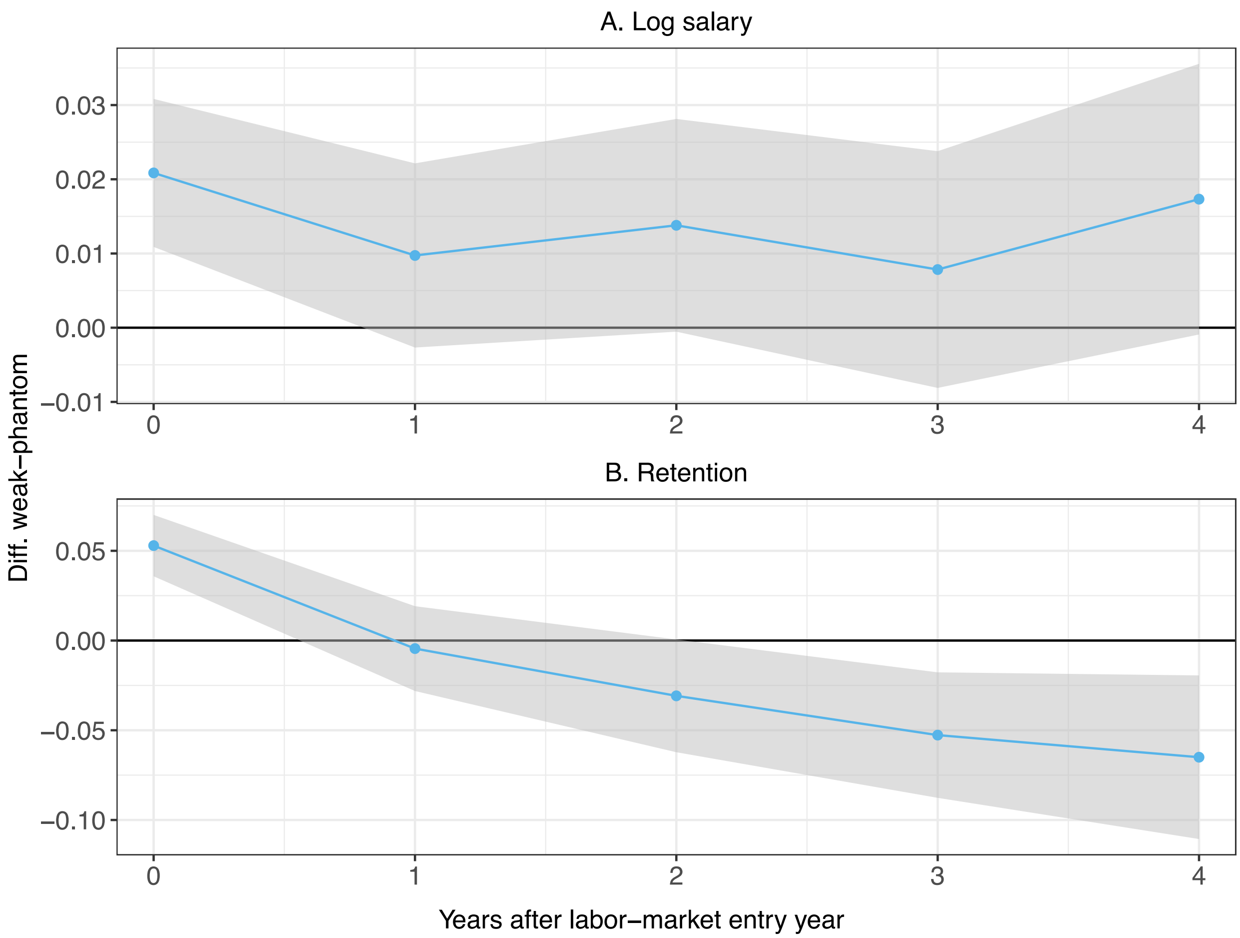


Notes: This figure shows the probability of working in a firm with weak connections (with one contact) for different values of firm and match characteristics, relative to the probability of working in a phantom-connected firm (with one contact). The left column (Panels A, C, and E) shows results by deciles of each characteristic across all firms. The right column (Panels B, D, and F) shows the correlation between the sector-level effect of weak connections and the mean value of each characteristic in that sector. Panels A and B show results for mean match duration. Panels C and D show results for standard deviation of match duration. Panels E and F show results for match effects. Mining and Other sectors are excluded from Panels B, D, and F due to small sample sizes. The y-axis shows the mean (and 95% confidence interval) of the differences between the coefficients of weak and phantom connections across 100 estimations of equation (1), using a 10 percent random sample of workers each time, separately by bin or sector. The employment outcome is scaled by 100, so results are in percentage points.

Figure 5: Event-study plot of coefficients: Log salary

Notes: This figure shows the event-study version of the correlation between parental connections at the first job and log salary as a function of the number of years between the contact's last year at the firm and the worker's labor market entry year. The figure shows the estimated coefficients (and 95 percent confidence intervals) of phantom and weak connections with one contact from equation (4). The vertical line between -1 and 0 indicates the transition from worker-firm pairs with phantom connections to pairs with weak connections. The outcome is log monthly salary in the first year of employment. Estimates are relative to workers with no connections at their first job. The regression includes group and firm fixed effects. Groups are constructed using all combinations of workers' observable characteristics (ethnicity, education, gender, year of first job, age, and district of residence). The 95 percent confidence intervals are based on standard errors clustered at the firm level.

Figure 6: Correlation between parental connections at first job and salary and job tenure over time



Notes: This figure shows the correlation between parental connections at the first job and salary and retention as a function of years since the labor-market entry year. The outcome in Panel A is log monthly salary, and in Panel B is retention (an indicator for staying at the firm in the next year). The regressions include first-firm and group fixed effects and control for calendar year, period (years since entry), and age fixed effects. Groups are constructed using all combinations of workers' observable characteristics (ethnicity, education, gender, year of first job, age, and district of residence). The graph shows the differences between the coefficients of weak and phantom connections (with one contact) and their 95% confidence intervals, constructed using robust standard errors clustered at the firm level.

Appendices

A Appendix tables and figures

Table A1: Summary statistics, firms

	1-4	5-500	501+
Firms	123,677	51,999	392
Workers	225,830	1,155,398	833,097
Av. firm size	1.83	22.23	2131.56
Share of firms	0.702	0.296	0.002
Share of workers	0.102	0.522	0.376

Notes: This table reports summary statistics for firms according to the number of workers in the firm. The first row shows the overall number of unique firms in the 2006-2015 matched employee-employer files. The second row shows the total number of workers in each group of firms by year, averaged across years. The third row shows the average number of workers in a firm by year, averaged across years. The fourth and fifth rows show the share of firms and the share of workers in each group of firms by year, averaged across years.

Table A2: Event-study plot of coefficients: Effect of parental connections on firm assignment

	Employment		
Phantom connections		Weak connections	
-5	0.005	0	0.058
	(0.002)		(0.003)
-4	0.005	1	0.052
	(0.001)		(0.004)
-3	0.007	2	0.042
	(0.001)		(0.006)
-2	0.009	3	0.040
	(0.002)		(0.006)
-1	0.012	4	0.050
	(0.002)		(0.009)
1	0.026	5	0.041
	(0.003)		(0.005)
2	0.017	Other connections	
	(0.002)		
3	0.013		0.490
	(0.002)		(0.008)
4	0.009		
	(0.002)		
5	0.009		
	(0.002)		

Notes: This table reports the mean (and standard deviation) of the estimated coefficients of parental connections across 100 estimations of equation (2) using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points. For both phantom and weak connections, the numbers indicate the last year the parent's coworker worked at the firm relative to the worker's labor-market entry year. Negative values indicate phantom connections (contact left before entry), while non-negative values (0-5) indicate weak connections (contact left at or after entry). "Other connections" includes firms where the worker has more than one weak and/or phantom connection, mixed connections (both weak and phantom), or strong connections.

Table A3: Correlation between parental connections and employment at first job by firm size

	Past firm size		Current firm size	
	Small	Large	Small	Large
	(1)	(2)	(3)	(4)
Phantom: 1	0.019	0.005	0.012	0.011
	(0.001)	(0.001)	(0.001)	(0.002)
Phantom: 2	0.062	0.020	0.041	0.030
	(0.006)	(0.005)	(0.004)	(0.010)
Phantom: 3+	0.058	0.025	0.041	0.038
	(0.010)	(0.007)	(0.007)	(0.014)
Weak: 1	0.084	0.013	0.052	0.032
	(0.003)	(0.002)	(0.002)	(0.005)
Weak: 2	0.503	0.039	0.339	0.068
	(0.037)	(0.012)	(0.025)	(0.033)
Weak: 3+	1.632	0.096	1.184	0.199
	(0.141)	(0.032)	(0.100)	(0.069)
Mixed (phantom/weak)	0.416	0.076	0.265	0.142
	(0.015)	(0.007)	(0.010)	(0.017)
Strong	6.386	1.801	6.571	2.373
	(0.116)	(0.165)	(0.125)	(0.221)
Diff. weak-phantom: 1	0.065	0.008	0.040	0.021
	(0.003)	(0.002)	(0.002)	(0.006)
Diff. weak-phantom: 2	0.442	0.019	0.299	0.038
	(0.038)	(0.012)	(0.026)	(0.035)
Diff. weak-phantom: 3+	1.573	0.072	1.143	0.161
	(0.142)	(0.032)	(0.101)	(0.074)
Observations	12,859,088	12,228,832	19,826,973	1,595,300

Notes: This table reports the mean (and standard deviation) of the estimated coefficients of parental connections on employment at the first job across 100 estimations of equation (1), separately by firm size, using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points. Columns 1-2 split the sample by past firm size (the size of the firm where the parent's coworker was employed): small firms have fewer than 200 employees and large firms have 200-500 employees. Columns 3-4 split the sample by current firm size (the size of the firm at the time of potential employment): small firms have fewer than 200 employees and large firms have 200-500 employees. Each connection type is categorized by the number of contacts: one, two, or three or more. "Diff. weak-phantom" reports the difference between the coefficients of weak and phantom connections for each category.

Table A4: Balancing test: Correlation between parental connections and measures of proximity between workers and firms

	Log distance	Parent's industry
	(1)	(2)
Phantom connections	-0.369	0.077
	(0.004)	(0.000)
Weak connections	-0.368	0.076
	(0.003)	(0.001)
Other connections	-0.926	0.281
	(0.009)	(0.001)
Diff. weak-phantom	0.001	-0.001
	(0.002)	(0.000)
R0 (no connections)	10.102	0.033
	(0.007)	(0.000)
Ratio weak-phantom	1.000	0.989
	(0.000)	(0.003)
Observations (firms x groups)	21,166,443	21,166,443
N firms	149,729	149,729
N groups	2,959	2,959
N workers	220,684	220,684

Notes: This table compares the geographical distance between a worker and a firm and the probability that a firm belongs to the same 3-digit industry as the worker's parent for firms with parental connections relative to non-connected firms. The table reports the mean (and standard deviation) of the estimated coefficients of phantom, weak, and other connections across 100 estimations of equation (1) with the outcome variables mentioned using a 20 percent random sample of workers each time. Phantom and weak connections include only firms with one contact. Other connections include firms where the worker has more than one weak and/or phantom connection, mixed connections (both weak and phantom), or strong connections. R0 is the average outcome variable's value for a non-connected firm. "Diff. weak-phantom" is the difference between the coefficients of weak and phantom connections. "Ratio weak-phantom" is the estimated odds ratio between the outcome variable's value for a weakly-connected firm and a phantom-connected firm.

Table A5: Effects of parental connections on firm assignment: death and retirement of contacts

	Employment		
	(1)	(2)	(3)
Special connections:	Death	Retirement	Death or retirement
Phantom (D/R)	0.031	0.010	0.017
	(0.016)	(0.010)	(0.008)
Phantom (non-D/R)	0.010	0.010	0.010
	(0.001)	(0.001)	(0.001)
Weak (D/R)	0.065	0.032	0.041
	(0.033)	(0.016)	(0.015)
Weak (non-D/R)	0.050	0.051	0.051
	(0.002)	(0.002)	(0.002)
Other	0.487	0.487	0.487
	(0.008)	(0.008)	(0.008)
R0 (no connections)	0.005	0.005	0.005
	(0.000)	(0.000)	(0.000)
Ratio weak-phantom (D/R)	2.567	3.913	2.773
	(2.219)	(5.313)	(3.861)
Ratio weak-phantom (non-D/R)	3.679	3.680	3.691
	(0.209)	(0.212)	(0.214)
N connections: phantom (D/R)	85,532	138,194	222,461
N connections: weak (D/R)	37,402	102,499	138,974

Notes: This table reports the mean (and standard deviation) of the estimated coefficients of phantom, weak, and other connections across 100 estimations of equation (1) with separate coefficients for "D/R" connections (death, retirement, or both, depending on the column) and "non-D/R" phantom and weak connections, using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points. "Death" connections are connections in which the contact died no more than one year after the last year she worked at the firm. "Retirement" connections are connections in which the last year the contact worked at the firm was at the mandatory retirement age (62 for females and 67 for males). Column 3 uses either death or retirement connections. Phantom and weak connections include only firms with one contact. "Other" includes firms where the worker has more than one weak and/or phantom connection, mixed connections, or strong connections. R0 is the average probability of working in a non-connected firm. "Ratio weak-phantom (D/R)" is the estimated odds ratio between working in a D/R weakly-connected firm and working in a D/R phantom-connected firm. "Ratio weak-phantom (non-D/R)" is defined similarly for non-D/R connections.

Table A6: Effect of weak parental connections on firm assignment, placebo test

	All	Jews	Arabs	Males	Females
	(1)	(2)	(3)	(4)	(5)
Phantom connections	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.001)	(0.000)	(0.001)
Weak connections	0.000	0.000	0.000	0.000	0.000
	(0.001)	(0.001)	(0.003)	(0.001)	(0.002)
Other connections	0.000	0.000	0.001	0.000	0.000
	(0.003)	(0.002)	(0.010)	(0.004)	(0.005)
Diff. weak-phantom	0.000	0.000	0.000	0.000	0.000
	(0.001)	(0.001)	(0.004)	(0.001)	(0.002)
R0 (no connections)	0.007	0.006	0.011	0.008	0.007
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Ratio weak-phantom	1.010	1.000	1.053	1.011	1.017
	(0.152)	(0.149)	(0.337)	(0.176)	(0.239)
Observations	21,166,443	16,837,526	4,328,917	15,319,313	5,847,130
N firms	149,729	144,186	117,746	145,939	134,555
N groups	2,959	1,658	1,301	1,548	1,411
N workers	220,684	157,009	63,675	170,872	49,812
N connections	40,827,833	33,261,814	7,566,019	31,664,340	9,163,493

Notes: This table shows placebo test results, assigning the worker's connections to a random worker in her group. The table reports the mean (and standard deviation) of the estimated coefficients of phantom, weak, and other connections across 100 estimations of equation (1) based on the new (randomized) data using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points. Phantom and weak connections include only firms with one contact. "Other" includes firms where the worker has more than one weak and/or phantom connection, mixed connections, or strong connections. "Diff. weak-phantom" reports the difference between the coefficients of weak and phantom connections. R0 is the average probability of working in a non-connected firm. "Ratio weak-phantom" is the estimated odds ratio between working at a weakly-connected firm and working at a phantom-connected firm. The first column reports results for the entire sample, while columns 2-3 report results by ethnicity (Jews and Arabs) and columns 4-5 report results by gender (Males and Females).

Table A7: Correlation between parental connections at first job and salary, job tenure, and wage growth by ethnicity

	Log salary		Job tenure		Salary growth	
	Jews	Arabs	Jews	Arabs	Jews	Arabs
	(1)	(2)	(3)	(4)	(5)	(6)
Phantom: 1	0.016	-0.001	0.072	0.100	0.016	-0.002
	(0.005)	(0.006)	(0.025)	(0.038)	(0.013)	(0.018)
Phantom: 2	0.006	0.031	0.005	0.314	0.055	0.045
	(0.010)	(0.012)	(0.060)	(0.084)	(0.032)	(0.031)
Phantom: 3+	0.005	0.023	0.154	0.294	0.055	0.032
	(0.015)	(0.015)	(0.068)	(0.126)	(0.041)	(0.037)
Weak: 1	0.031	0.017	0.193	0.158	-0.001	-0.040
	(0.005)	(0.006)	(0.030)	(0.034)	(0.015)	(0.017)
Weak: 2	0.052	0.035	0.336	0.340	-0.020	0.015
	(0.017)	(0.013)	(0.094)	(0.084)	(0.036)	(0.038)
Weak: 3+	0.095	0.047	0.461	0.249	0.027	-0.012
	(0.023)	(0.015)	(0.118)	(0.075)	(0.051)	(0.066)
Mixed (phantom/weak)	0.031	0.041	0.265	0.252	-0.007	0.002
	(0.006)	(0.006)	(0.034)	(0.036)	(0.014)	(0.017)
Strong	0.135	0.089	0.639	0.508	0.031	-0.007
	(0.005)	(0.005)	(0.033)	(0.042)	(0.012)	(0.012)
Diff. weak-phantom: 1	0.015	0.018	0.121	0.058	-0.017	-0.038
	(0.007)	(0.008)	(0.040)	(0.048)	(0.021)	(0.023)
Diff. weak-phantom: 2	0.047	0.004	0.332	0.026	-0.076	-0.030
	(0.019)	(0.020)	(0.111)	(0.117)	(0.048)	(0.050)
Diff. weak-phantom: 3+	0.089	0.024	0.307	-0.045	-0.028	-0.044
	(0.028)	(0.021)	(0.132)	(0.142)	(0.065)	(0.076)
Mean (no connections)	8.600	8.511	1.904	2.014	0.338	0.238
Observations	157,023	63,783	157,023	63,783	77,340	29,028

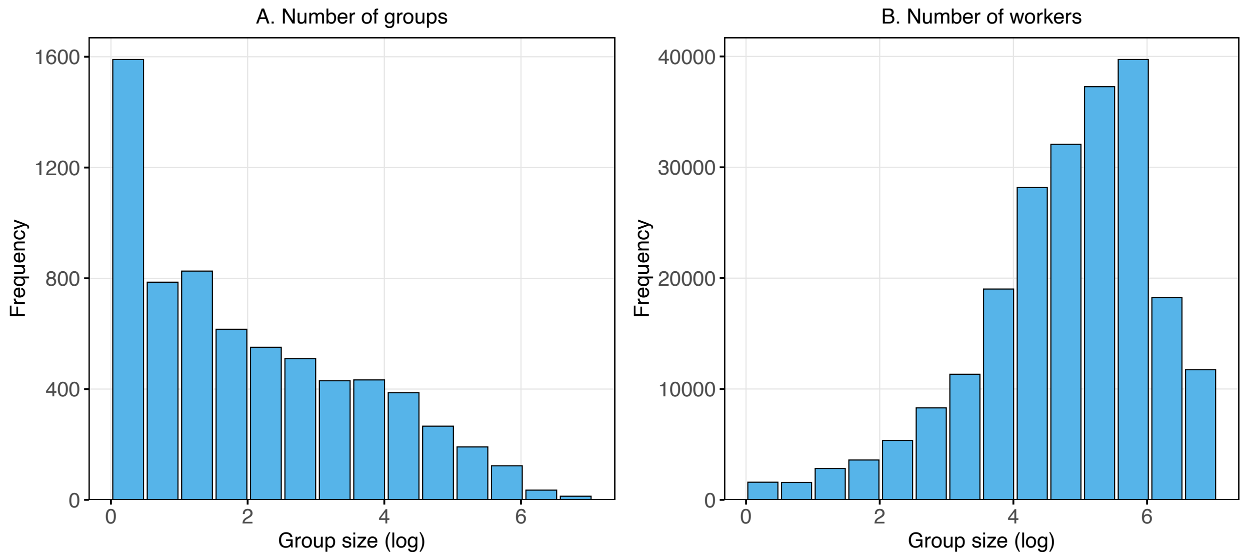
Notes: This table reports the correlation between parental connections and log salary (columns 1-2), job tenure (columns 3-4), and salary growth (columns 5-6), separately for Jewish and Arab workers. See Table 4 notes for detailed variable definitions. All specifications include group fixed effects and firm fixed effects.

Table A8: Correlation between parental connections at first job and salary, job tenure, and wage growth by gender

	Log salary		Job tenure		Salary growth	
	Male	Female	Male	Female	Male	Female
	(1)	(2)	(3)	(4)	(5)	(6)
Phantom: 1	0.008	0.016	0.096	0.061	0.010	-0.001
	(0.005)	(0.006)	(0.028)	(0.034)	(0.014)	(0.016)
Phantom: 2	0.011	0.009	0.109	0.063	0.015	0.115
	(0.011)	(0.012)	(0.064)	(0.078)	(0.030)	(0.041)
Phantom: 3+	0.015	0.016	0.306	0.055	0.035	0.065
	(0.014)	(0.018)	(0.082)	(0.101)	(0.038)	(0.047)
Weak: 1	0.028	0.013	0.183	0.172	-0.020	-0.001
	(0.005)	(0.007)	(0.030)	(0.039)	(0.015)	(0.019)
Weak: 2	0.040	0.067	0.408	0.223	-0.016	0.022
	(0.012)	(0.020)	(0.075)	(0.116)	(0.034)	(0.047)
Weak: 3+	0.058	0.072	0.236	0.558	-0.011	0.021
	(0.015)	(0.026)	(0.068)	(0.173)	(0.055)	(0.061)
Mixed (phantom/weak)	0.039	0.024	0.294	0.155	0.004	-0.019
	(0.006)	(0.006)	(0.033)	(0.041)	(0.013)	(0.018)
Strong	0.131	0.089	0.689	0.338	0.025	-0.004
	(0.005)	(0.007)	(0.035)	(0.040)	(0.011)	(0.017)
Diff. weak-phantom: 1	0.020	-0.003	0.087	0.111	-0.029	0.000
	(0.007)	(0.008)	(0.039)	(0.052)	(0.021)	(0.023)
Diff. weak-phantom: 2	0.028	0.058	0.299	0.159	-0.030	-0.093
	(0.017)	(0.022)	(0.099)	(0.138)	(0.047)	(0.060)
Diff. weak-phantom: 3+	0.043	0.056	-0.070	0.503	-0.046	-0.044
	(0.021)	(0.032)	(0.104)	(0.195)	(0.066)	(0.076)
Mean (no connections)	8.636	8.503	1.933	1.933	0.340	0.278
Observations	126,233	94,573	126,233	94,573	62,636	43,732

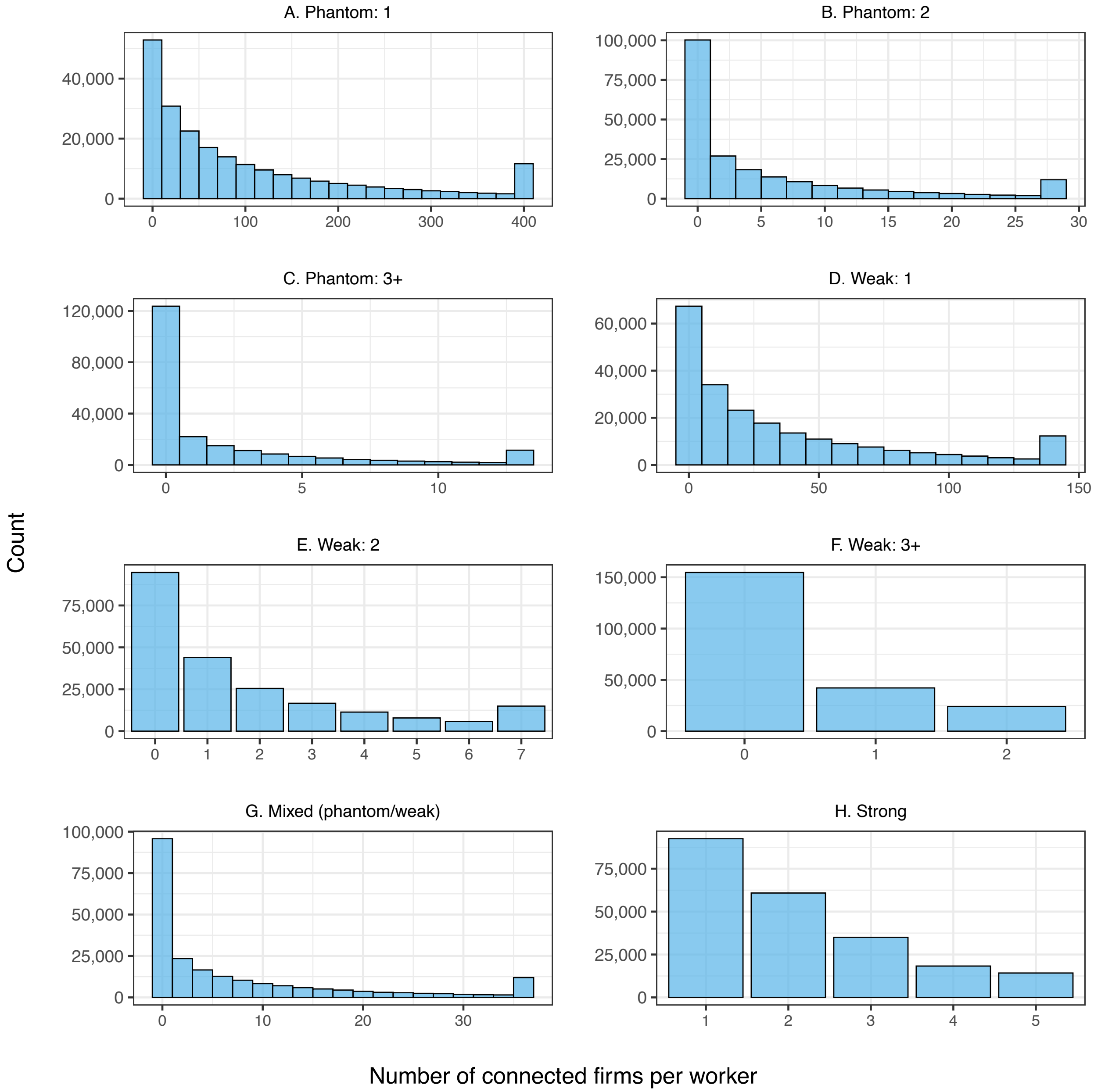
Notes: This table reports the correlation between parental connections and log salary (columns 1-2), job tenure (columns 3-4), and salary growth (columns 5-6), separately for Male and Female workers. See Table 4 notes for detailed variable definitions. All specifications include group fixed effects and firm fixed effects.

Figure A1: Distribution of group size



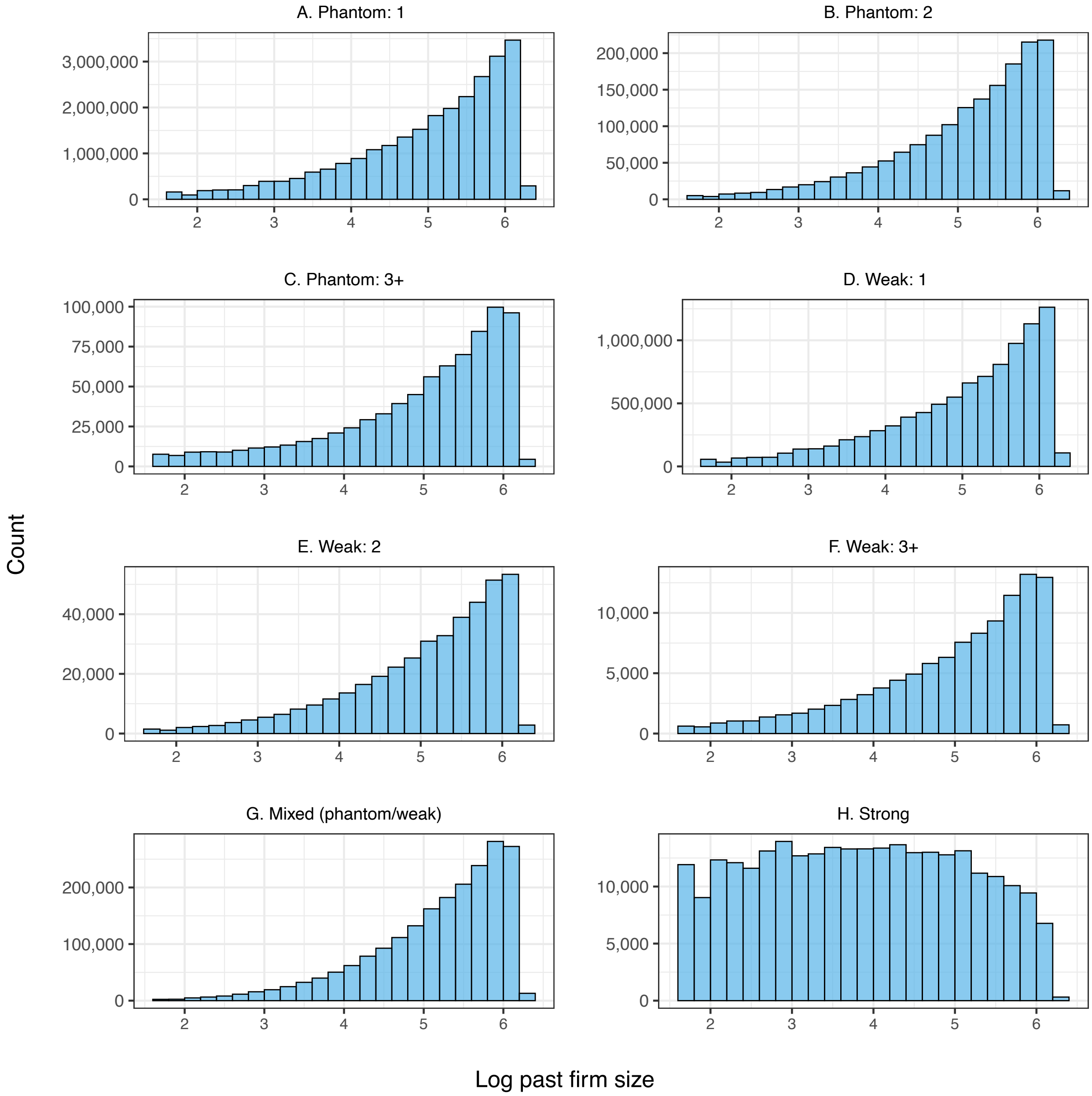
Notes: This figure shows the distribution of group sizes in the sample. Groups are constructed using all combinations of the workers' observable characteristics (ethnicity, education, gender, year of the first job, age, and district of residence). Panel A shows the number of groups in each bin. Panel B shows the number of workers, calculated by multiplying the number of groups by the average group size in each bin. The x-axis shows group size in logarithmic scale.

Figure A2: Distribution of connections per worker by connection type



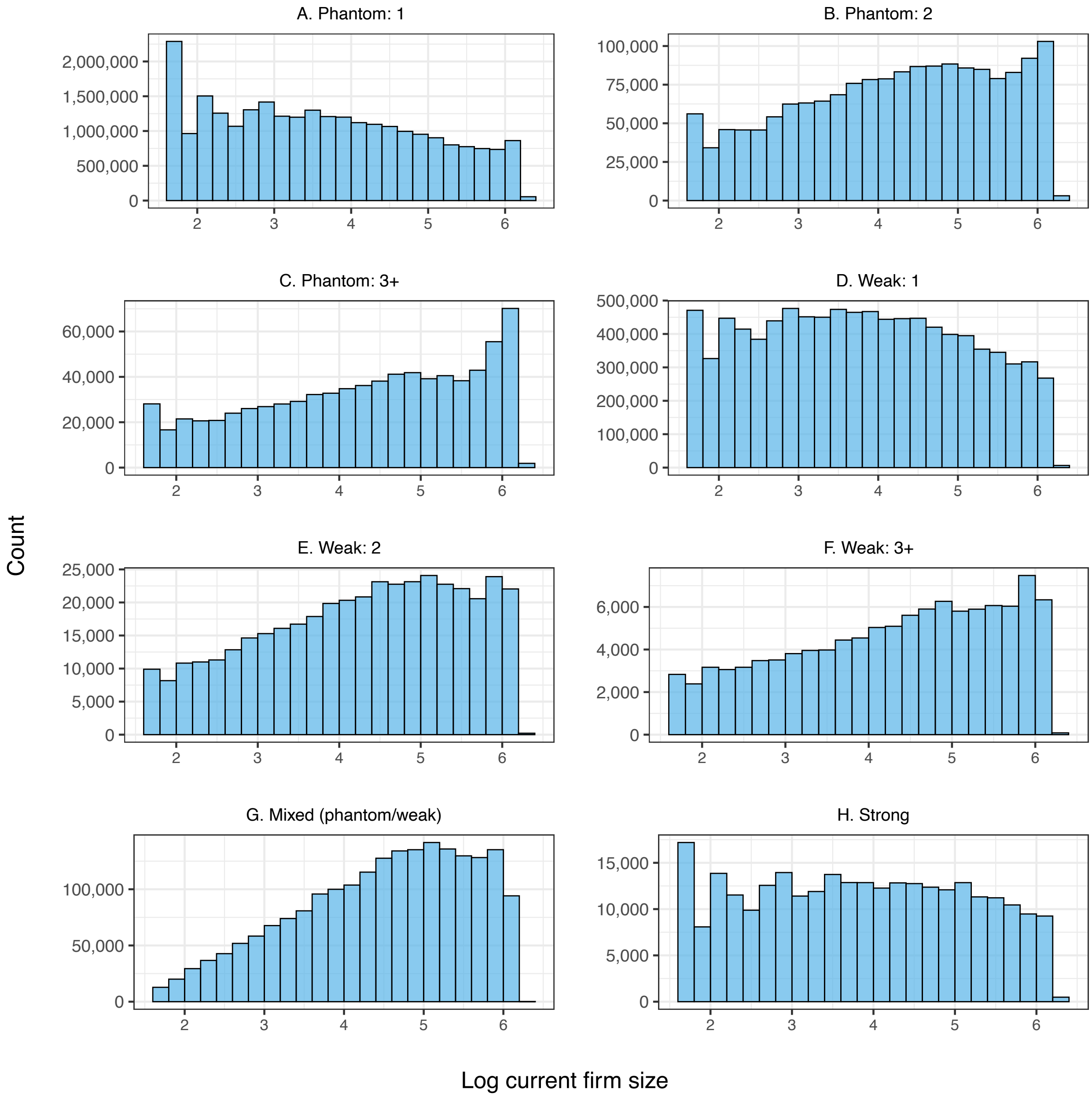
Notes: This figure shows the histogram of the number of connected firms per worker by connection type. The last bin in each panel includes workers with connections equal to or greater than the displayed value.

Figure A3: Number of worker-firm connections by past firm size and connection type



Notes: This figure shows the Number of worker-firm connections by past firm size (in logs) and connection type. Each panel shows a histogram for one of the eight connection types.

Figure A4: Number of worker-firm connections by current firm size and connection type

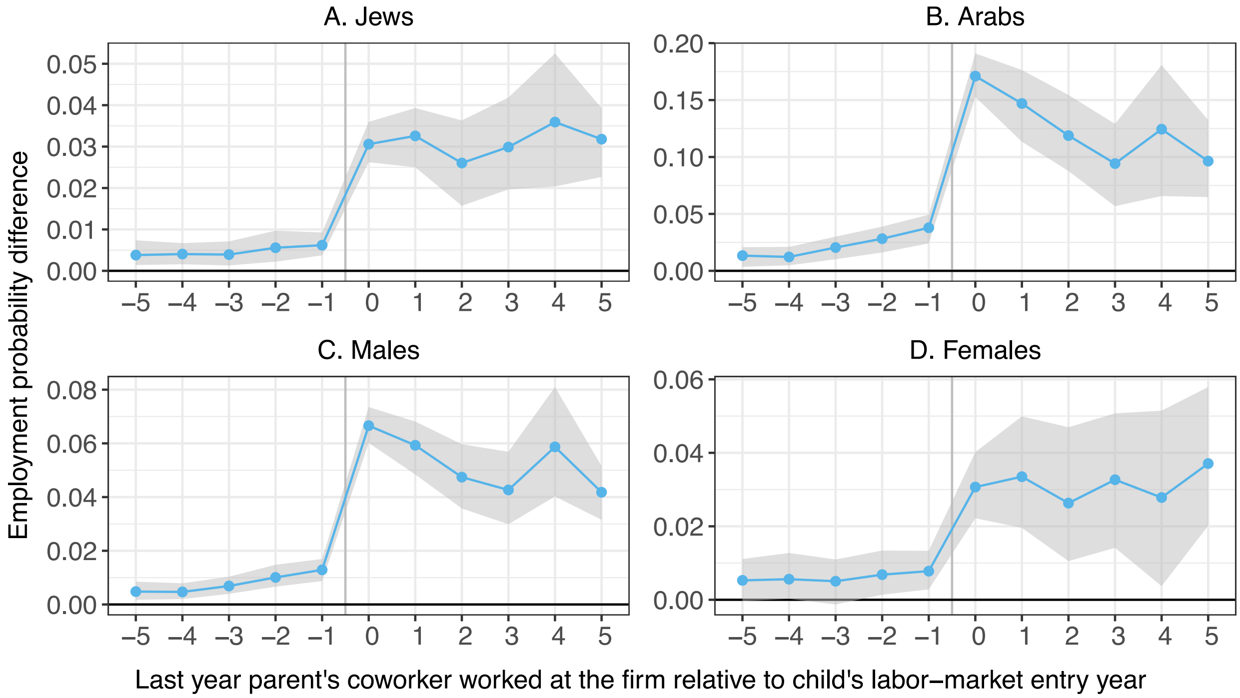


Notes: This figure shows the Number of worker-firm connections by current firm size (in logs) and connection type. Each panel shows a histogram for one of the eight connection types.

Figure A5: Raw data: Probability of working in a firm for phantom and weak connections

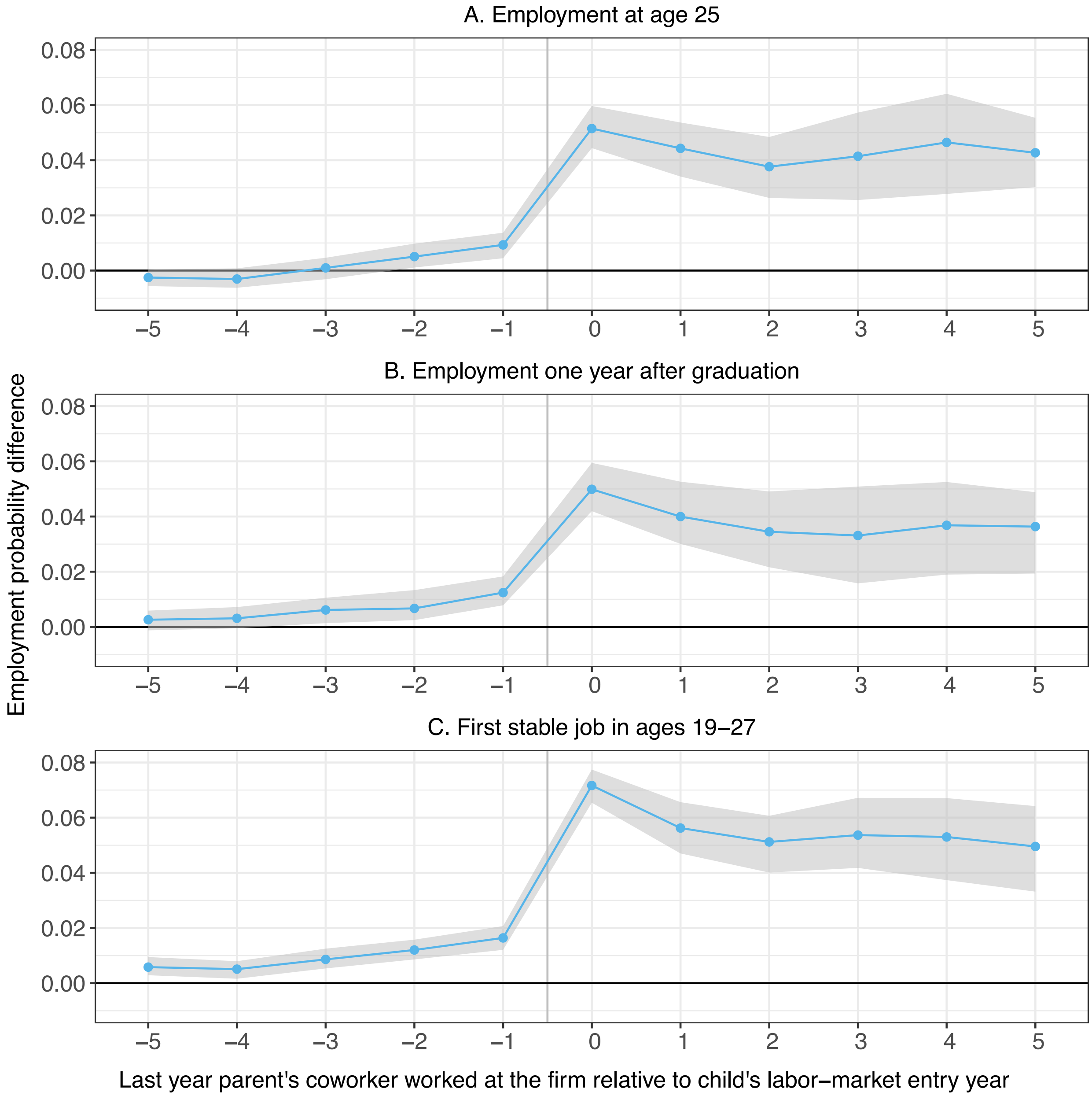
Notes: This figure shows the raw probability of working in a firm as a function of the difference between the last year the parent's coworker worked at the firm and the worker's labor-market entry year. The figure includes only connections with one contact. The employment outcome is scaled by 100, so results are in percentage points. The vertical line between -1 and 0 indicates the change from worker-firm pairs with phantom connections to pairs with weak connections.

Figure A6: Event-study plot of coefficients: Effect of weak parental connections on firm assignment (by groups of workers)



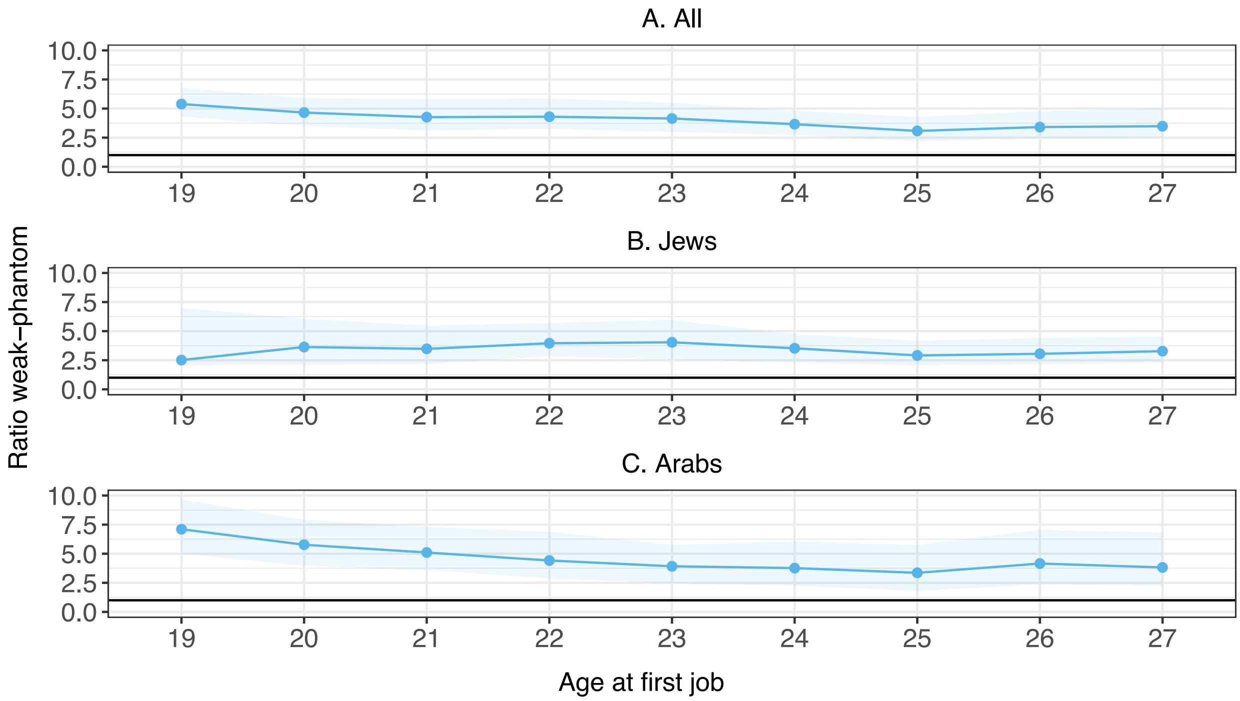
Notes: This figure shows the probability of working in a firm as a function of the difference between the last year the parent's coworker worked at the firm and the worker's labor-market entry year, relative to working in a non-connected firm, separately for sub-groups of workers. Phantom and weak connections include only firms with one contact. The figure shows the mean (and 95 percent confidence interval) of the estimated coefficients of phantom and weak connections across 100 estimations of equation (2) using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points.

Figure A7: Event-study plot of coefficients: Effect of weak parental connections on firm assignment (alternative definitions of the labor-market entry year)



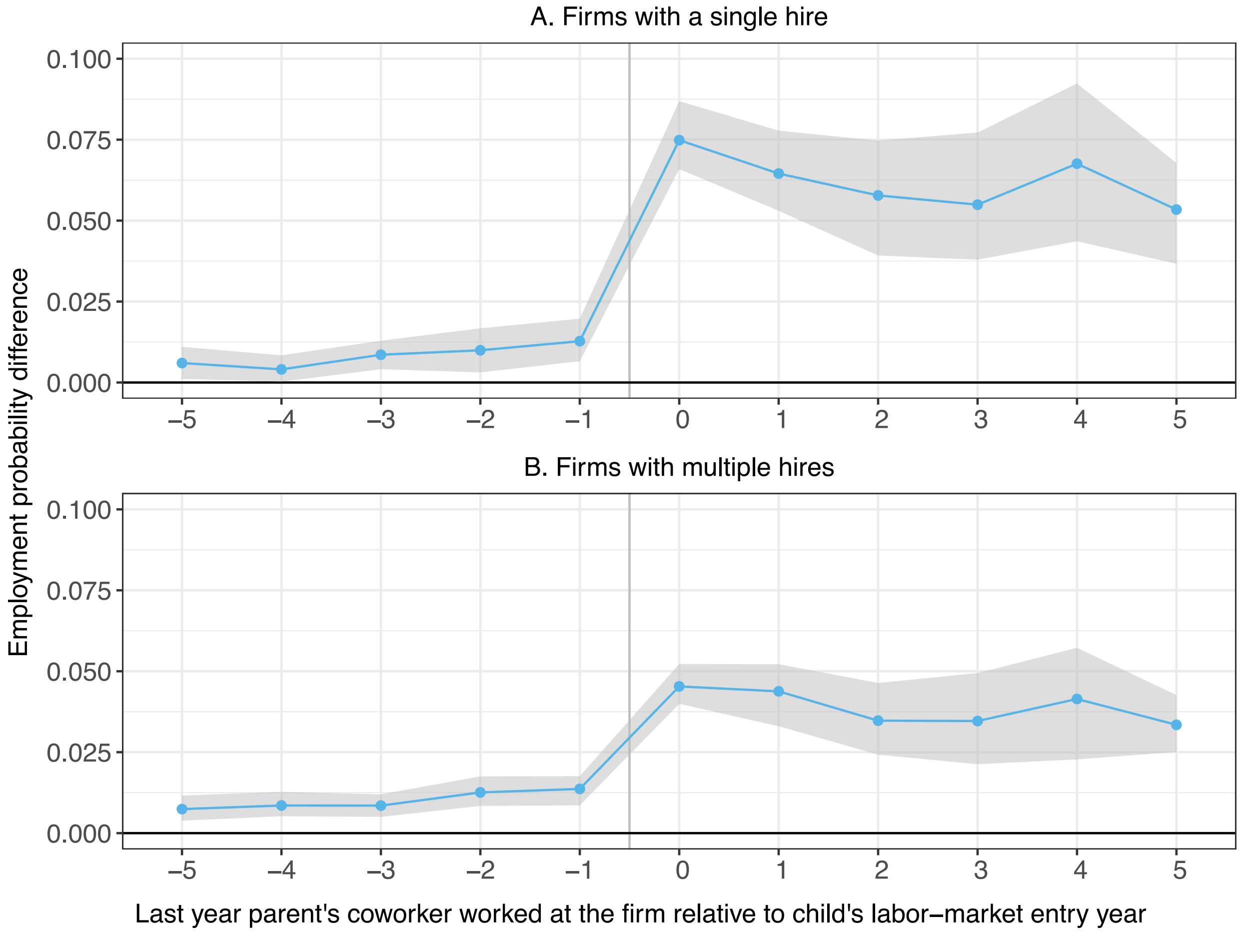
Notes: This figure shows the probability of working in a firm as a function of the difference between the last year the parent's coworker worked at the firm and the worker's labor-market entry year, for alternative definitions of labor-market entry. Phantom and weak connections include only firms with one contact. Panel A uses age 25 as entry year, Panel B uses the year after graduation, and Panel C uses ages 19-27 (instead of 22-27). Results show the mean (and 95 percent confidence interval) across 100 estimations of equation (2) using a 20 percent random sample each time. The employment outcome is scaled by 100, so results are in percentage points.

Figure A8: Effect of weak parental connections on firm assignment as a function of age at first job and ethnicity



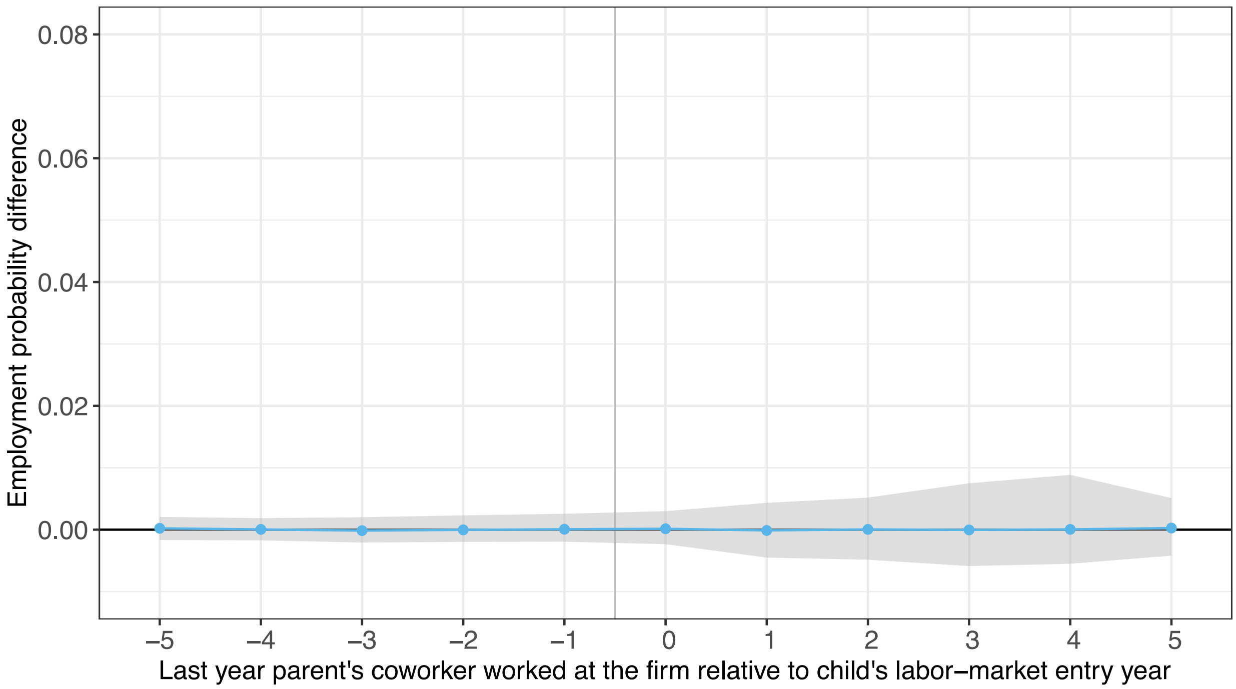
Notes: This figure shows the ratio between the probability of working in a firm with weak connections and phantom connections as a function of the age at first job. Weak and phantom connections include only firms with one contact. The figure shows the mean (and 95 percent confidence interval) of the estimated ratio across 100 estimations of equation (2) using a 20 percent random sample of workers each time.

Figure A9: Event-study plot of coefficients: Effect of weak parental connections on firm assignment for firms with single and multiple hires



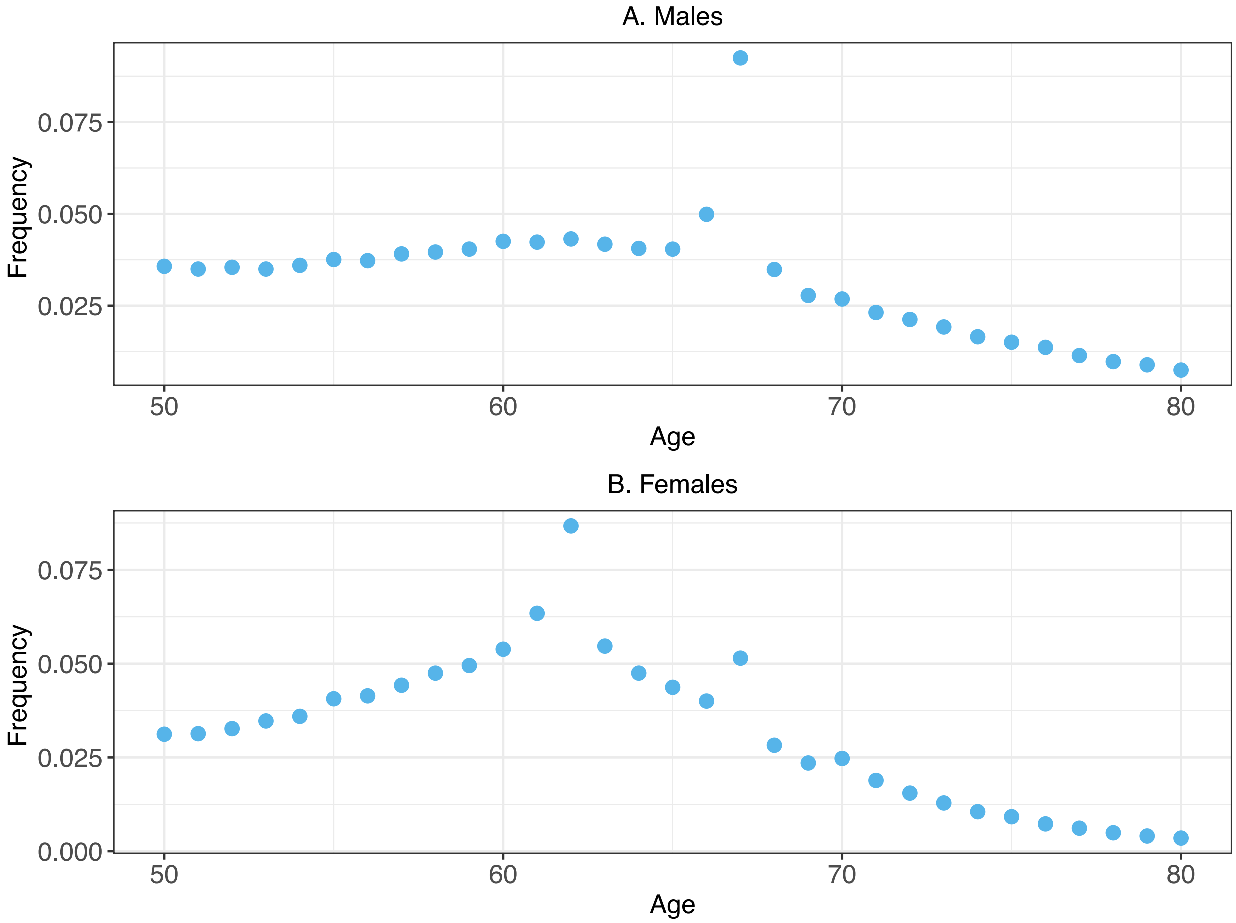
Notes: This figure shows the probability of working in a firm as a function of the difference between the last year the parent's coworker worked at the firm and the worker's labor-market entry year relative to working in a non-connected firm, for firms that hire single and multiple new workers per year. Phantom and weak connections include only firms with one contact. All panels show the mean (and 95 percent confidence interval) of the estimated coefficients of phantom and weak connections across 100 estimations of equation (2) using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points.

Figure A10: Event-study plot of coefficients: Effect of weak parental connections on firm assignment, placebo test



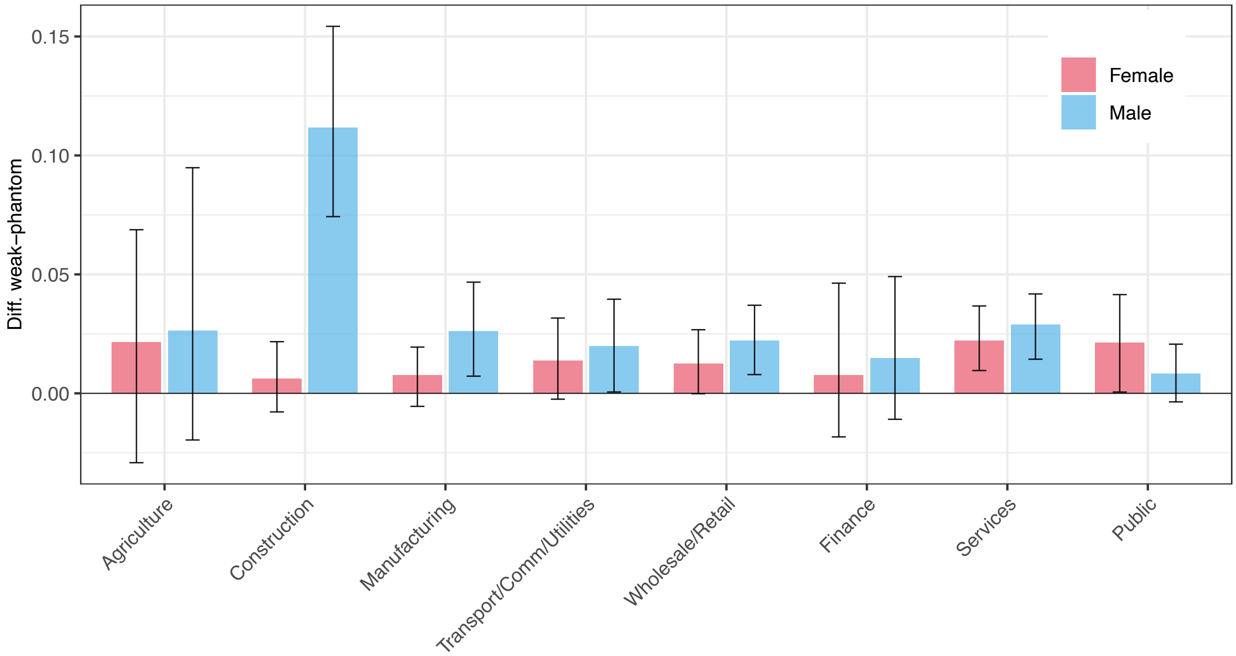
Notes: This figure reports the results of a placebo test, assigning the worker's connections to a random worker in her group. Phantom and weak connections include only firms with one contact. The figure shows the probability of working in a firm as a function of the difference between the last year the parent's coworker worked at the firm and the worker's labor-market entry year, relative to the probability of working in a non-connected firm, based on the new (randomized) data. The points are the mean coefficients of phantom and weak connections across 100 estimations of equation (2) using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points. The bounds of the 95 percent confidence intervals are constructed using the 2.5 and 97.5 percentiles of the coefficients' distribution.

Figure A11: Age at last year of work by gender



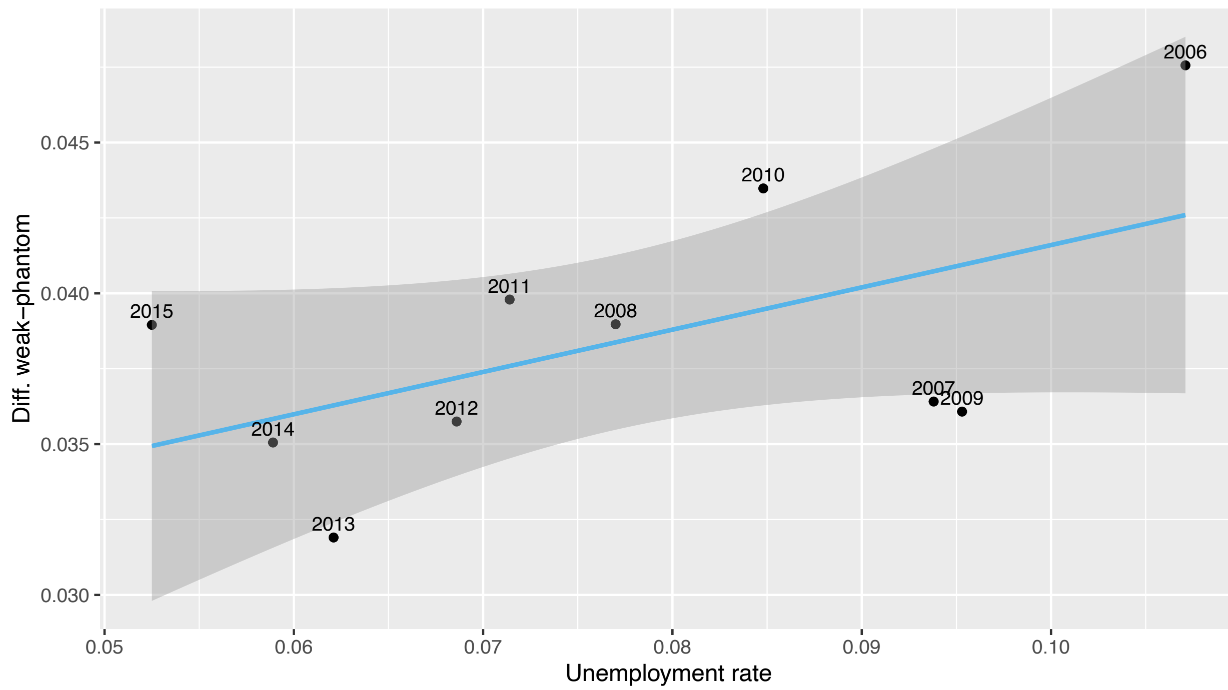
Notes: This figure shows the frequency of the ages of workers when they last appear in the employer-employee data between 2006-2014, separated by gender. Workers that worked in 2015 (the final year in the dataset) are not included in this figure. The figure includes workers that were between 50-80 in their last year of work.

Figure A12: Effects of weak parental connections on firm assignment by sector and gender



Notes: This figure shows the probability of working in a firm with weak connections for different sectors, separately by gender, relative to the probability of working in a phantom-connected firm. Weak and phantom connections include only firms with one contact. The bars show the mean differences between the coefficients of weak and phantom connections across 100 estimations of equation (1), using a 20 percent random sample of workers each time. The employment outcome is scaled by 100, so results are in percentage points. The bounds of the 95 percent confidence intervals are constructed using the 2.5 and 97.5 percentiles of that distribution of the differences. Mining and Other sectors are excluded from the figure due to the small number of observations.

Figure A13: Correlation between the effects of weak parental connections on firm assignment and total unemployment rate by year



Notes: The vertical axis is the probability of working in a weak-connected firm relative to the probability of working in a phantom-connected firm by labor-market entry year. Weak and phantom connections include only firms with one contact. The horizontal axis is the total unemployment rate of the total labor force in that year. The coefficients (and robust standard errors) of the fitted line are 0.028 (0.007) and 0.14 (0.089) for the intercept and slope, respectively. The Pearson correlation coefficient is 0.55.

B Data appendix

This appendix provides additional details on data preparation and variable definitions.

Employment and wages: The data contain observations at the worker $\times$ firm $\times$ year level. Each observation includes monthly employment indicators and total yearly salary. I cleaned the data by (1) dropping observations with missing worker or firm identifiers, (2) replacing empty monthly indicators with zeros, (3) dropping observations that are duplicated across all variables, and (4) for duplicate worker-firm observations, taking the maximum of the monthly indicators and the sum of yearly earnings.

Parents and children: The data include the identifiers of the mother and father of each Israeli citizen, provided that they are also Israeli citizens.

Education: Starting in 1996, I observe the higher-education institution and period of enrollment for each individual in Israel. Workers without any period of enrollment in higher education institutions are defined as "no college" workers. Workers with at least one year of admission to higher education institutions (excluding religious schools) are defined as workers with "some college" or "college" education.

Ethnicity: Workers are classified into two categories: Arabs and Jews. Arabs include Arab Muslims, Arab Christians, Druze, and Circassians. Following the practice of the Israeli Central Bureau of Statistics, I include "Jews and Others" together in the definition of Jews and consider workers without ethnicity classification as Jews.[1]

Workers' location: I measure the worker's residence location by the longitude and latitude of the centroid of the city she lived in at age 21. I also use the worker's district at age 21, one of seven districts: North, Haifa, Tel Aviv, Center, Jerusalem, South, and Judea and Samaria.

Natives: Individuals born in Israel with no information on the date of immigration.

Ultraorthodox: I use the internal algorithm of the National Insurance Institute, which uses information on residency neighborhoods, educational institutions, and family links to identify Ultraorthodox individuals.

Industry: I clean the industry variable such that each firm has a unique industry. Using the employer-employee file described above with additional information on the 4-digit industry code of the firm in each observation, for each year I: (1) drop observations with missing worker, firm, or industry identifiers, and (2) for each firm, keep the industry with the most occurrences. If the number of firms in industry A in year t that changed their

[1] According to Israeli Central Bureau of Statistics definitions, "Others" refers to Non-Arab Christians, members of other religions, and those not classified (CBS 2019). The majority of people in this category are immigrants from the former Soviet Union who immigrated to Israel in the past three decades. They are not Jews according to Jewish law but are included in the Law of Return because of their familial ties with Jews (Cohen and Susser 2009).

industry in year $t+1$ to B is greater than the number of firms that remain in industry A, I assume the classification of that industry changed and update industry A to B retroactively. Finally, for each firm, I keep the most recent industry. In practice, I use the implied 3-digit industry code for each firm (2011 Israeli classification).

Firms' location: Exact information on firm locations is missing. As a proxy, I calculate the median longitude and latitude of workers' residences. I exclude new workers from the calculation of firm locations.[2]

Panel data construction: I construct a panel dataset at annual frequency. Following Kramarz and Skans (2014), I assign each person-year observation to the firm in which that person was employed during February. I calculate monthly salary by dividing yearly salary at a firm by the number of months worked there. If someone worked at more than one firm during February, I assign them to the firm that paid the higher monthly salary. I exclude from the sample worker-year observations with earnings less than 25% of the national average monthly wage.[3] The sample period is 1991-2015, and I keep workers aged 22-69 each year.[4] I construct a second dataset from this panel, keeping only firms with 5-500 workers each year.[5] I use these data to build the parental network over time and the sample of new workers (see below).

First stable job and labor-market entry year: Following Kramarz and Skans (2014), I define a stable job as a job after higher-education graduation (if applicable) that lasts at least four months during a calendar year and produces total annual earnings of at least 150% of the national average monthly wage. The first stable job is the first job satisfying the conditions above for workers aged 22-27 at a firm with 5-500 workers. The labor-market entry year is the year the worker finds her first stable job.[6]

[2]The data do not include establishment/plant identifiers or an indicator for multi-plant firms. Therefore, I assign the same location to all branches or plants of the same firm. This problem is alleviated by dropping firms with more than 500 workers from the sample.

[3]The minimum monthly salary in 2015 was 48.8% of the average salary in that year. This ratio fluctuated between 40%-50% during 1991-2015. Therefore, I exclude workers who earn approximately 50% or less of the minimum wage, similar to Kramarz and Skans (2014).

[4]Many people in Israel serve in the military between ages 18-21, especially Jews, so I exclude these ages to prevent military service differences from affecting the estimated wage gaps between groups. However, I include ages 19-21 in robustness checks reported at the end of Section C.3.

[5]The first reason for this restriction is computational: including large firms adds a huge number of potential connections. Additionally, incorporating large firms introduces significant noise into the estimation. Intuitively, the probability that a random pair of workers forms genuine social connections decreases as firm size increases. In Figure 3 Panel A, I show that the effect of having a parental connection at a firm on the probability of working at that firm decreases as firm size increases. Moreover, this effect disappears for firms with more than 400 workers. Therefore, assuming that worker pairs in large firms have social connections would increase measurement error in connections and could potentially bias the effect of connections downward. During 2006-2015, firms with 5-500 workers accounted for 29.6% of firms and employed 52.2% of workers (Table A1).

[6]In Section 3.8, I study the effects of parental connections on being employed. I also check the robustness

New workers: My analysis sample of new workers comprises the first stable job of Israelis aged 22-27 at firms with 5-500 workers who found this job during 2006-2015. I exclude workers without any parent who worked at a firm with 5-500 workers when the worker was aged 12-21. I further exclude immigrants and Ultraorthodox Jews from the sample.[7]

Firms' pay premium: Firm pay premiums are estimated using the AKM model (Abowd et al. 1999):

$$w_{it} = \alpha_i + \psi_{J(it)} + Z'_{it}\gamma + \varepsilon_{it}, \tag{B1}$$

where w_{it} is the log monthly salary of worker i in year t, α_i is a person fixed effect, $\psi_{J(it)}$ is a firm fixed effect, and Z'_{it} is a set of year fixed effects and a cubic age function restricted to be flat at age 40 (Card et al. 2018). I exclude new workers so their salaries do not impact the estimated firm premiums. To capture potential changes in a given firm's premium over the years, I estimate a separate regression for each year. Specifically, firm premiums at year t are calculated using the largest connected set from the full sample in years $[t-4, t]$. Finally, I rank the estimated firm premiums within each year ("firm rank").[8]

Weak connections: A worker i has weak connections to a firm j if i's parent and a worker k worked simultaneously at a firm $j' \neq j$ when i was aged 12-21, and k worked at firm j at i's labor-market entry year. Both past and current firms employ between 5 and 500 employees. I study separately weak connections with one, two, or three or more contacts.

Phantom connections: A worker i has phantom connections to a firm j if i's parent and a worker k worked simultaneously at a firm $j' \neq j$ when i was aged 12-21, and k worked at firm j at any time within five years before or after i's labor-market entry year, but not that year. I study separately phantom connections with one, two, or three or more contacts.

Strong connections: A worker i has strong connections to a firm j if i's parent worked at firm j when i was aged 12-21.

Mixed connections: A worker i has mixed connections to a firm j if i has both weak and phantom connections to firm j.

of the main results to alternative definitions of the labor-market entry year (see Appendix C.3).

[7]Immigrants often do not have parental connections in the labor market. See Arellano-Bover and San (2023) for the role firms play in explaining pay gaps between former Soviet Union immigrants and natives in Israel. Ultraorthodox Jews in Israel have unique labor-market characteristics, such as low secular education and employment rates, especially for males (Berman 2000; Fuchs and Epstein 2019). Specific research is needed to study this group.

[8]These premiums aim to capture the average differences in salaries that firms pay to similar workers. They are not necessarily a proxy for firm productivity but might capture other factors that lead to salary differences, such as differential rent sharing and firm pay policy (Card et al. 2018; Amior and San 2025). See Card et al. (2018) for a discussion of the AKM model and its critique.

Other connections: A worker i has other connections to a firm j if i has: (1) weak or phantom connections with more than one contact, (2) mixed connections, or (3) strong connections to firm j.[9]

C Regression appendix

C.1 Econometric model: identification

To ease notation, in what follows, I assume there is only one (comparable) type of weak and phantom connections (e.g., both with one contact only). Analogous analysis can be done for additional types (e.g., with two or three or more contacts).

Assume the true model is:

$$e_{ij} = \phi_{x(i)j} + \sum_{c=p,w} \delta^{true,c} \cdot D^c_{ij} + \nu_{ij} + u_{ij}. \tag{C1}$$

with $E(u_{ij}|D^w_{ij}, D^p_{ij}, x(i) \times j) = 0$. ν_{ij} is the (unobserved) part in the error that is not orthogonal to connections (given the fixed effects).

The main identification assumption is[10]:

$$E(\nu_{ij}|D^w_{ij}, x(i) \times j) = E(\nu_{ij}|D^p_{ij}, x(i) \times j). \tag{C2}$$

I also assume that the true effect of phantom connections is zero:

$$\delta^{true,p} = 0 \tag{C3}$$

ν_{ij} is unobserved, so in practice, the estimation equation is:

$$e_{ij} = \phi_{x(i)j} + \sum_{c=p,w} \delta^{c} \cdot D^c_{ij} + \epsilon_{ij}. \tag{C4}$$

For simplicity, from now on, I abstract from the fixed effects $\phi_{x(i)j}$.

From the Omitted Variable Bias (OVB) formula (e.g., Angrist and Pischke (2009)), the

[9]To reduce endogeneity in measuring connections, I define the parent's past firms and past coworkers using a fixed time period (when the child is aged 12-21). I do not include connections formed in the years between when the child is 22 and when they enter the labor market. Doing so would mechanically increase the set of connections available for workers who enter the labor market later.

[10]Denoting $\epsilon_{ij} = \nu_{ij} + u_{ij}$, this condition is equivalent to the condition in the text $E(\epsilon_{ij}|D^w_{ij}, x(i) \times j) = E(\epsilon_{ij}|D^p_{ij}, x(i) \times j)$, as, by the Law of Iterated Expectations, $E(u_{ij}|D^w_{ij}, x(i) \times j) = 0$ and $E(u_{ij}|D^p_{ij}, x(i) \times j) = 0$.

estimate of δ^w equals[11]:

$$\delta^w = \delta^{true,w} + \frac{cov(D^w, \nu)}{var(D^w)} \tag{C5}$$

Similarly, and using Assumption C3, we have:

$$\delta^p = \frac{cov(D^p, \nu)}{var(D^p)} \tag{C6}$$

Now, with some algebra, we get

$$\frac{cov(D^w, \nu)}{var(D^w)} = E(\nu|D^w = 1) - E(\nu|D^w = 0) \tag{C7}$$

and:

$$\frac{cov(D^p, \nu)}{var(D^p)} = E(\nu|D^p = 1) - E(\nu|D^p = 0) \tag{C8}$$

But because of Assumption C2, we have $E(\nu|D^p = 1) = E(\nu|D^w = 1)$ (and obviously $E(\nu|D^p = 0) = E(\nu|D^w = 0)$). Therefore:

$$\delta^{true,w} = \delta^w - \delta^p. \tag{C9}$$

Three comments are noteworthy here. First, I support the main identification assumption (C2) in Sections 3.4 and 3.5. Specifically, in Table A4 (see Appendix C.4), I show that regressing observed variables, which are likely to increase the probability of employment, on indicators for weak and phantom connections yields identical coefficients for both types of connections.

Second, Assumption C3 is strong and cannot be tested in my framework. As mentioned

[11] Note that I obtain the OVB formula of a single independent variable, although I have two independent variables, as they are mutually exclusive binary variables. To see this, assume the model is $y_i = b_0 + b_1 x_1 + b_2 x_2 + \epsilon$. The OLS estimate of b_1 is:

$$\hat{b}_1 = \frac{(\sum x_2^2)(\sum x_1 y) - (\sum x_1 x_2)(\sum x_2 y)}{(\sum x_1^2)(\sum x_2^2) - (\sum x_1 x_2)^2}$$

Because x_1 and x_2 are binary mutually exclusive variables, we have $x_1 x_2 = 0$, so

$$\hat{b}_1 = \frac{\sum x_1 y}{\sum x_1^2}$$

which is the formula for the OLS estimate of a single independent variable. The same argument holds for $\hat{b}_2$.

in the text, if the true effect of phantom connections is positive, my identification strategy provides a lower bound for the true effect of weak connections.

Third, my identification strategy fits well with weak connections (separately with different numbers of contacts), and that is the main focus of this paper. I also report the estimates for mixed and strong connections, but the results should be interpreted with caution given the absence of an appropriate control group.

C.2 Econometric model: computation

Starting with equation (1), let $D_{ij} \equiv \max_c \left[D_{ij}^c\right]$ be a variable that indicates whether worker i has any type of connections at firm j. First, I restrict the sample under study to cases in which there is within group-firm variation in D_{ij}. This restriction has no impact on the parameters of interest since the discarded observations are uninformative conditional on the fixed effects (Kramarz and Skans 2014). I then aggregate the model by computing, for each group-firm combination, the fraction of workers with connections at the firm that this firm hired:

$$R_{xj}^{CON} = \frac{\sum_{i\in x} e_{ij} D_{ij}}{\sum_{i\in x} D_{ij}} = \phi_{xj} + \sum_c \delta^c \cdot D_{xj}^c + \epsilon_{xj}^{CON} \tag{C10}$$

where $D_{xj}^c = \frac{\sum_{i\in x} D_{ij}^c}{\sum_{i\in x} D_{ij}}$ is the share of c-type connections for workers in group x who are connected to firm j. Similarly:

$$R_{xj}^{-CON} = \frac{\sum_{i\in x} e_{ij}(1-D_{ij})}{\sum_{i\in x}(1-D_{ij})} = \phi_{xj} + \epsilon_{xj}^{-CON} \tag{C11}$$

Taking the difference between the two ratios eliminates the firm-group fixed effects ϕ_{xj}:

$$R_{xj} \equiv R_{xj}^{CON} - R_{xj}^{-CON} = \sum_c \delta^c \cdot D_{xj}^c + \epsilon_{xj}^R. \tag{C12}$$

The variable R is computed for each firm-group combination as the fraction of hires at the firm from the group having any type of connection to that firm minus the fraction of hires at the firm from the same group having no parental connection to that firm. The right-hand side variables D_{xj}^c capture the fraction of connected workers from group x who have the specific connection type c to firm j. The estimates of δ^c from equation (C12) measure the effect of the different types of parental connections.[12]

[12]Note that, by definition, $\sum_c D_{xj}^c = 1$, which means that the independent variables in equation (C12) are collinear. However, estimation of that equation is feasible because the regression is estimated without an intercept.

Even after the fixed-effects transformation, limited computational resources prevent estimation of the model using all observations. Therefore, I take a 20 percent random sample of the new workers in each iteration and run 100 such iterations. Using the distribution of estimates obtained, I calculate the mean and 95 percent confidence intervals of the regression coefficients and other statistics of interest.

C.3 Robustness checks

Following Kramarz and Skans (2014), I define the labor-market entry year as the year the worker finds her first job. This definition may be endogenous if social connections affect not only the identity of the firm a worker finds but also the time until she finds a stable job (or her decision to start looking for jobs). I check the robustness of the results to the definition of the labor-market entry year by estimating the effect of connections using two alternative definitions: (1) the year in which the worker is 25 years old, and (2) the year after the worker's graduation year. Figure A7, Panels A and B, plots the event-study coefficients using these alternative definitions. Overall, the results look very similar to the benchmark results, with a discrete increase in the probability of employment at time zero of about 0.04 percentage points.

Next, in the analysis so far, I have defined the first stable job as employment found between ages 22-27. This definition was chosen because many people in Israel, especially Jews, serve in the military between ages 19-21, and I wanted to avoid these differences affecting the estimated wage gaps between groups. However, in Figure A7 Panel C, I allow people to find their first stable job also at ages 19-21, and the results remain similar. Furthermore, I re-estimated the main specification as a function of entry age. The effect is indeed stronger for younger cohorts, particularly for Arabs (see Figure A8).

Finally, since I do not observe departments or divisions within firms, in the main analysis I assume that if a worker has connections of a specific type to a firm, she has connections to all relevant jobs at that firm. Figure A9 presents the event-study plot for firms with single and multiple hires separately. Similar patterns exist in both cases, although the effect is stronger for firms with single hires. The differential effects might result from fewer spillovers in firms with single hires and could also reflect a decrease in measurement error.

C.4 Balancing test

As mentioned earlier, social connections between a worker and a firm might be correlated with other similarity measures between the worker and the firm. Two leading examples are the geographical distance between the worker and the firm and the similarity between the firm and the firms at which the worker's parents have worked. Indeed, in what follows, I show that the distance between workers and firms is smaller if there are parental connections between the worker and the firm. Likewise, the probability that the firm is in the same industry as one of the parent's firms is higher if there are connections. In the first test of the identification strategy, I check whether there are also such differences between phantom and weak parental connections.

To do so, I re-estimate equation (1) with the distance/similarity measures as the outcome variable. Weak and phantom connections include only firms with one contact. The first measure is the distance between the worker's location at age 21 and the firm's location.[13] Column 1 of Table A4 shows the estimated coefficients. As expected, compared to firms with no connections, firms with all types of social connections are closer to workers' locations. However, the estimates for phantom and weak connections are virtually identical, at -0.369 and -0.368 log points.

The second measure is an indicator variable that equals one if the firm has the same 3-digit industry code as one of the parents' previous firms. Once again, connected workers were more likely to have parents who worked in the same industry than unconnected workers. This correlation, however, is similar for phantom and weak connections, with estimates of 0.077 and 0.076 percentage points, respectively (Table A4, column 2).

C.5 Exogenous separation: death and retirement of potential contacts

This paper's identification strategy exploits the timing of parents' coworkers' employment relative to workers' labor-market entry. I assume that other than the effect of social connections at the time of job search, there is no systematic difference in the probability of working at weak- and phantom-connected firms. This assumption breaks if the separation time is correlated with other factors unrelated to social connections that affect employment decisions. For example, workers who leave a firm might deliver to their contacts a negative opinion about the possibility of working at that firm. This mechanism would decrease the probability of working at the firm only for workers whose contacts left the firm before they started to work, not after. In this case, having phantom connections at the firm would decrease the job seeker's probability of working there compared to weak connections.

To investigate this possibility further, I estimate the effect of connections for two exogenous reasons for separations. The first specific separation cause is death. I classify the separation cause as "death" if the contact died no more than one year after working at the firm. I compare the probability of working at firms where the (dead) potential contact worked

[13] I do not use the worker's location at the labor-market entry year to avoid the mechanical correlation between workers' locations and the firm resulting from moving closer to the workplace.

at the firm before time zero to the probability of working at firms in which the connection worked at the firm after that time and died immediately afterward.

The second separation cause is quitting the job precisely at retirement age. During the years of analysis, the statutory retirement age in Israel was 62 for females and 67 for males. At that age, workers are entitled to leave their jobs and receive a pension. Figure A11 plots the distribution of workers' ages in the last year of employment for males and females. This figure shows that it is common to leave the labor force at retirement age. I compare workers who quit their firm at retirement age, before and after year zero.

For each special type of connection, I split the set of phantom and weak connections into two subsets, each with connections belonging to the death/retirement group (i.e., the contact died or left the job at retirement age) and connections that do not belong to that group. Phantom and weak connections include only firms with one contact. I then re-estimate equation (1) using the five types of connections (phantom-death/retirement, phantom-non-death/retirement, weak-death/retirement, weak-non-death/retirement, and other).

Table A5 reports the results of this exercise. Compared to fresh graduates without connections to the firm, the probability of working at the firm with a contact who died while employed at the firm or immediately afterward is higher by 0.031 percentage points if the last year the contact worked at the firm was before time zero and by 0.065 percentage points if it was after time zero. The estimates for firms with non-death/retirement contacts, i.e., contacts who did not die in the year after leaving the firm, are virtually identical to the baseline results for weak and phantom connections with one contact in Table 2 (0.01 and 0.05 for phantom and weak connections, respectively). The ratio between the probability of working at a firm with weak connections compared to a firm with phantom connections is 2.6 for death/retirement connections and 3.7 for non-death/retirement connections (Table A5, column 1). However, due to the small number of such cases, the estimated ratio for death/retirement connections is not statistically significantly different from 1. Similar results were obtained when using the statutory retirement age as a special case of job separation. Once again, the estimates for firms with contacts who left the firm exactly at retirement age are higher for weak connections than phantom connections (0.032 and 0.010 percentage points, respectively). The ratio of weak to phantom connections is 3.9 for connections at firms where the contacts left at retirement age, compared to 3.7 for non-death/retirement connections. I also estimate the effect by combining the death and retirement causes of separation. The estimated ratio between weak and phantom connections is 2.8, compared to 3.7 for non-death/retirement connections. Again, these ratios are not statistically significantly different from 1 (Table A5, columns 2 and 3).

Overall, the estimated effects of connections are quantitatively similar for contacts who left the firm for exogenous reasons (death or retirement) and non-death/retirement contacts. The ratio between the probability of working at a firm with weak connections and a firm with phantom connections is slightly smaller for death and somewhat larger for retirement than for non-death/retirement connections. However, due to the small number of connections belonging to these types, the estimates of the special types of connections are much noisier and not statistically different from zero. These results suggest (with the aforementioned caveat) that the estimated effects of connections obtained from the benchmark model (with all connections) are not a result of endogenous separation that differentially impacts phantom and weak connections but rather the effects of the connections themselves.

C.6 Placebo test: assigning worker's connections to another worker

Another threat to the identification strategy is if firms with different types of connections have different hiring trends. For example, suppose connections leave (become "phantoms") when demand for a particular type of labor is falling. In that case, the firms that usually hire this type of labor will hire fewer new workers regardless of the impact of connections.

To address this concern, I perform a placebo test and assign a worker's connections to another worker in her group. If the employment probability gap between actual- and phantom-connected firms is mediated by other factors correlated with the different types of connections, the probability of a worker working at a firm that another group member has real connections to will be higher than at a firm that another group member has phantom connections to. On the other hand, if the employment probability is higher only if the connections are the worker's true connections (and not the connections of someone else with similar observable characteristics), that suggests that the estimated effect is the effect of the connections themselves.

Table A6 reports the estimates of equation (1), assuming each worker has the set of connections of a random member of her group. Phantom and weak connections include only firms with one contact. None of the estimates are statistically significantly different from zero. Moreover, there is no statistically significant difference between the estimated probability of working at a weak-connected firm and a phantom-connected firm. The event-study estimates of equation (2) also showed no difference between phantom and real-connected firms (Figure A10).[14]

[14] The fact that the estimated effect is not different from zero for the phantom-placebo connections is expected because the control group ("no connections") only includes worker-firm pairs in which someone from the worker's observable group has some type of connections at the firm (see the discussion after equation (1)). Still, if active (weak) links reflected a more positive employment trend at a firm than phantom links, we would see an increase in the employment probability for active connections (of someone else).

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